Simulation Analysis

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2025-04-19

Introduction and Overview

*I need to write this once I figure out what goes here

Data Generating Process

The simulations are a fully factored experiment.

Block Level Fixed Covariates
Number of Blocks: 5; ATE: .2; ICC: .2; Average Number of Observations Per Cluster: 20; Proportion of Clusters Treated: .5 (this is fixed across blocks); Number of District Level Covariates: 0; How much does the average cluster size vary across blocks (coefficient of variation)?: 0

Block Level Varying Covariates (Factors that are varied across simulations)

Three groups of block level parameters Blocks all the same, Blocks all messy, and Blocks all reasonable

Blocks all the same: How much do number clusters per block vary (coefficient of variation)?: 0 (none); ATE SD: 0; Correlation Between Treatment Effect and Cluster Size: 0; Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters: 0; Correlation of Proportion of Clusters Treated per Block and Impact: 0; Correlation Between Outcome Intercept and Cluster Size: 0

Blocks all messy: How much do number clusters per block vary (coefficient of variation)?: .85 (big); ATE SD: .6; Correlation Between Treatment Effect and Cluster Size: .8; Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters: .8; Correlation of Proportion of Clusters Treated per Block and Impact: .8; Correlation Between Outcome Intercept and Cluster Size: .8

Blocks all reasonable: How much do number clusters per block vary (coefficient of variation)?: .33 (average); ATE SD: .1; Correlation Between Treatment Effect and Cluster Size: .2; Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters: .1; Correlation of Proportion of Clusters Treated per Block and Impact: 0; Correlation Between Outcome Intercept and Cluster Size: .1

Each of the rest of these factors don't vary as a group but vary each individually and are fully factored with all other factors:

Average Number of Clusters per Block: 7,10,16

Cluster Level Varying Parameters

Correlation of Impact and Size of Cluster: 0, .8

Cluster Level Treatment Effect Standard Deviation: 0, .3

Cluster Level ICC: 0, .2

Number of Observations Per Cluster Varies: True, False

How much does the size of clusters vary (coefficient of variation)?: 0, .33, .6

Number of School-Level Covariates: 0, 1, 5

[School Level Covariate R Squared is always .5]

What does the bias look like for each estimator

*The next thing to do is to move all the simulation questions I'm answering to my emails and also to attach the page numbers to the analysis

Main Bullet Points:

*The estimators theory says are unbiased are indeed consistently unbiased (or have very low bias across the full range of simulations)

*The LRapw−FIpw−db has bias as large as -.025, which seems from the statistical model to be driven by the interactions of a large treatment effect standard deviation across clusters, high variability in cluster size, and correlations between treatment effect and cluster size. This is consitent with theory.

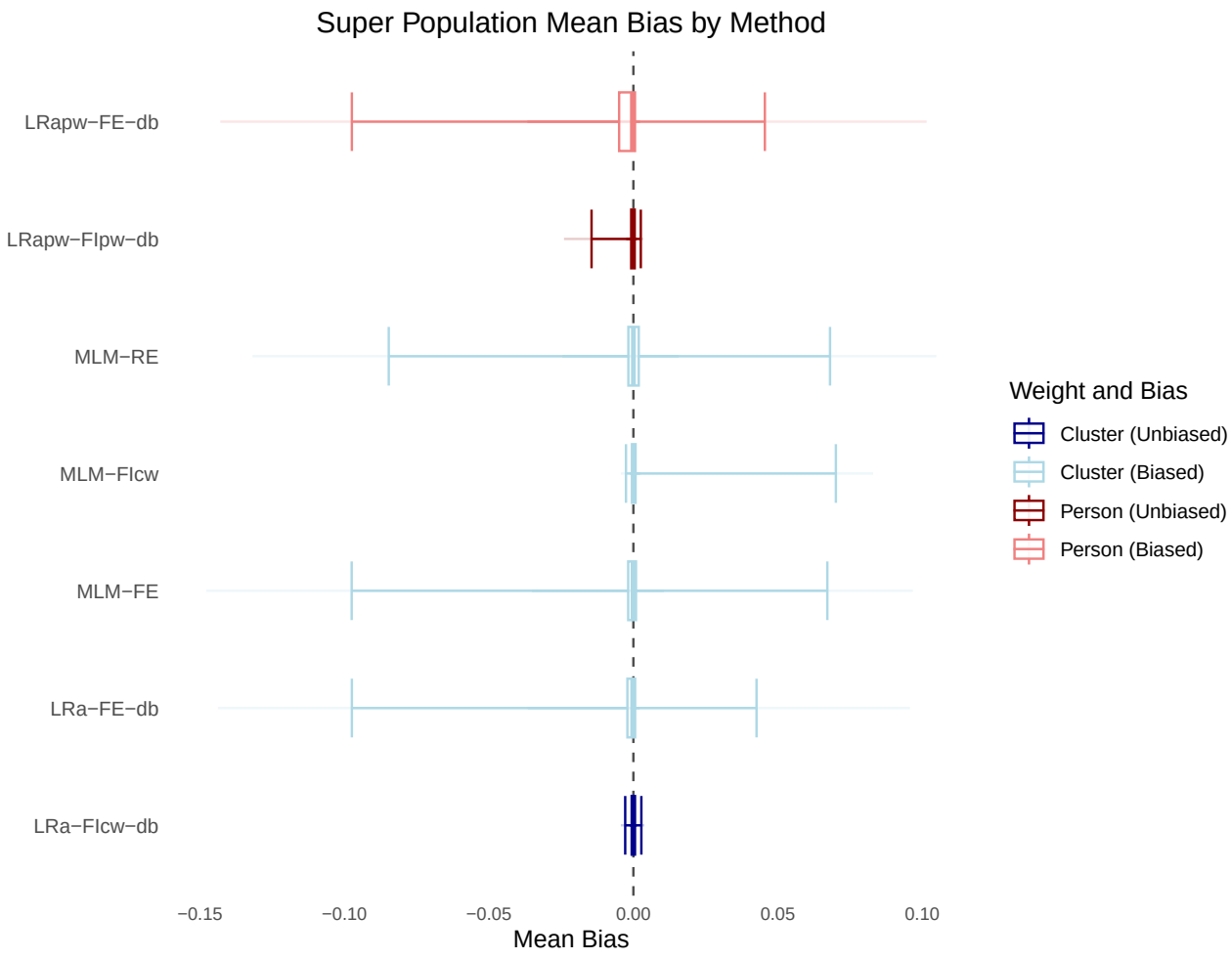
*For the various estimators that are biased, the mean bias is mostly driven by the different blocks being different from each other

Description of Analysis

*The graph is a boxplot of mean bias across the different scenarios by method. The faint horizontal lines go out to the full range, the bars are the 1% to 99%, and the boxes are the 10% to 90%.

*The first table contains the regression coefficients from a regression that predicts the magnitude of the mean bias using different data generating parameters. Each regression is run separately by estimator and mean bias is the mean bias against the targeted estimator.

*The second table contains the DGPs that produce that largest magnitude of mean bias for each method, conditional on the magnitude of the mean bias being at least .1 . Where again mean bias is the mean bias against the targeted estimator.



The Effect of Different Data Generating Parameters on the Magnitude of Mean Bias

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	.007.	<.001*	.009*	.002***	.009*	.005**	.009*
block_typeS	<.001	<.001	<.001	<.001	<.001	<.001	.001
block_typeL	.036***	<.001	.037***	<.001	.042***	.001	.036***
J.barS	-.003	<.001	-.003	<.001	-.004	-.001	-.005
J.barL	-.002	<.001	-.002	.001**	-.004	-.002	-.003
size.impact.cor.schoolL	-.005	<.001	-.006*	-.003***	-.009**	-.008***	-.008**
sd.schoolL	<.001	<.001	-.002	-.002***	-.005	-.008***	-.007*
ICC.schoolL	-.007*	.001***	-.008*	.001**	<.001	.010***	<.001
variable.nTRUE	-.006.	<.001	-.007*	-.002***	-.010**	-.008***	-.009**
cv.schoolS	-.006	<.001	-.006.	-.001***	-.008.	-.004*	-.008*
cv.schoolL	-.001	<.001	-.002	-.002***	-.005	-.009***	-.007.
num.cov.schoolS	-.005	<.001	-.005	<.001	-.005	<.001	-.004
num.cov.schoolL	.001	<.001	.001	<.001	<.001	<.001	.001
block_typeS:J.barS	<.001	<.001	<.001	<.001	<.001	<.001	-.002
block_typeL:J.barS	<.001	<.001	<.001	<.001	-.001	<.001	.001
block_typeS:J.barL	<.001	<.001	<.001	<.001	<.001	<.001	-.003
block_typeL:J.barL	-.001	<.001	<.001	<.001	-.001	<.001	.004
block_typeS: size.impact.cor.schoolL	<.001	<.001.	<.001	<.001	<.001	<.001	<.001
block_typeL: size.impact.cor.schoolL	-.006**	<.001	-.007**	<.001	-.011***	<.001	-.009***
block_typeS:sd.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:sd.schoolL	-.002	<.001	-.002	<.001	-.007**	<.001	-.007**

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
block_typeS:ICC.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	.002
block_typeL:ICC.schoolL	-.005*	<.001	-.004*	<.001	-.003	<.001	-.005*
block_typeS: variable.nTRUE	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: variable.nTRUE	.003	<.001	.001	<.001	-.002	<.001	-.002
block_typeS:cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:cv.schoolS	.003	<.001	.002	<.001	.001	<.001	<.001
block_typeS:cv.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:cv.schoolL	.003	<.001	.001	<.001	-.003	<.001	-.004
block_typeS: num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: num.cov.schoolS	-.003	<.001	-.003	<.001	-.003	<.001	-.002
block_typeS: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	.001
block_typeL: num.cov.schoolL	-.001	<.001	-.002	<.001	-.001	<.001	.001
J.barS: size.impact.cor.schoolL	.002	<.001	.002	<.001	.003	.001	.002
J.barL: size.impact.cor.schoolL	.003	<.001	.002	-.001***	.005.	.002	.004
J.barS:sd.schoolL	<.001	<.001	-.001	<.001	.001	.001	.002
J.barL:sd.schoolL	.001	<.001	<.001	-.001***	.003	.001	.003
J.barS:ICC.schoolL	.003	<.001	.003	<.001	.003	<.001	.002
J.barL:ICC.schoolL	.002	<.001*	.002	<.001	<.001	-.001	<.001
J.barS:variable.nTRUE	.003	<.001	.002	<.001	.004	.001	.003
J.barL:variable.nTRUE	.003	<.001	.002	-.001***	.005.	.001	.004.
J.barS:cv.schoolS	.002	<.001	.002	<.001	.002	.001	.003
J.barL:cv.schoolS	-.001	<.001	-.001	<.001	<.001	.001	.001
J.barS:cv.schoolL	-.002	<.001	-.002	<.001	-.001	.001	.001
J.barL:cv.schoolL	-.003	<.001.	-.004	-.001***	-.001	.001	<.001
J.barS:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	-.001
J.barS:num.cov.schoolL	-.003	<.001	-.003	<.001	-.003	<.001	-.002
J.barL:num.cov.schoolL	-.003	<.001	-.002	<.001	-.004	<.001	-.005.
size.impact.cor.schoolL: sd.schoolL	.003	<.001	.005**	.003***	.010***	.011***	.010***
size.impact.cor.schoolL: ICC.schoolL	.002	<.001	.002	<.001	-.002	-.006***	-.002
size.impact.cor.schoolL: variable.nTRUE	<.001	<.001*	.002	.003***	.007***	.011***	.007***
size.impact.cor.schoolL: cv.schoolS	.002	<.001	.003	.001***	.005*	.005***	.005*
size.impact.cor.schoolL: cv.schoolL	.001	<.001	.003	.002***	.008**	.011***	.007**
size.impact.cor.schoolL: num.cov.schoolS	.001	<.001	.001	<.001	.001	<.001	.001
size.impact.cor.schoolL: num.cov.schoolL	-.001	<.001	-.001	<.001	<.001	<.001	-.001
sd.schoolL:ICC.schoolL	<.001	<.001	<.001	<.001	-.005*	-.006***	-.005**
sd.schoolL:variable.nTRUE	.001	<.001	.003	.003***	.008***	.011***	.008***
sd.schoolL:cv.schoolS	-.003	<.001	-.002	.001***	<.001	.005***	.002
sd.schoolL:cv.schoolL	-.002	<.001	<.001	.002***	.006*	.011***	.008***
sd.schoolL: num.cov.schoolS	.001	<.001	.001	<.001	.001	<.001	.002
sd.schoolL: num.cov.schoolL	-.002	<.001	-.002	<.001	-.001	<.001	<.001
ICC.schoolL: variable.nTRUE	.004*	<.001	.004*	<.001	-.001	-.006***	-.001
ICC.schoolL:cv.schoolS	.005*	<.001	.005*	<.001	.003	-.003**	.002
ICC.schoolL:cv.schoolL	.002	<.001	.002	<.001	-.003	-.006***	-.003
ICC.schoolL: num.cov.schoolS	.002	<.001	.003	<.001	.002	<.001	.002
ICC.schoolL: num.cov.schoolL	.001	<.001	.001	<.001	<.001	<.001	.001

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
variable.nTRUE:cv.schoolS	.001	<.001	.002	.001***	.004.	.004***	.005*
variable.nTRUE:cv.schoolL	.002	<.001.	.003	.003***	.009***	.011***	.010***
variable.nTRUE: num.cov.schoolS	<.001	<.001	.001	<.001	.001	<.001	<.001
variable.nTRUE: num.cov.schoolL	.002	<.001	.001	<.001	.002	<.001	<.001
cv.schoolS: num.cov.schoolS	.005.	<.001	.005.	<.001	.005.	<.001	.005.
cv.schoolL: num.cov.schoolS	.002	<.001	.002	<.001	.002	<.001	.002
cv.schoolS: num.cov.schoolL	.002	<.001	.002	<.001	.002	<.001	.001
cv.schoolL: num.cov.schoolL	.002	<.001	.001	<.001	.003	<.001	.002

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Most Bias by Method

method	bias.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	-0.144	0	0	0	L	L	0	L
LRa-FE-db	-0.143	S	0	L	S	S	0	L
LRa-FE-db	-0.14	0	L	0	S	L	L	L
LRa-FE-db	-0.126	0	0	0	L	L	0	L
LRa-FE-db	-0.123	0	L	L	L	L	0	L
LRapw-FE-db	-0.143	S	0	L	S	S	0	L
LRapw-FE-db	-0.14	0	L	0	S	L	L	L
LRapw-FE-db	-0.134	0	0	0	L	L	0	L
LRapw-FE-db	-0.126	0	0	0	L	L	0	L
LRapw-FE-db	-0.114	0	L	0	0	0	0	L
MLM-FE	-0.148	0	0	0	L	L	0	L
MLM-FE	-0.143	S	0	L	S	S	0	L
MLM-FE	-0.14	0	L	0	S	L	L	L
MLM-FE	-0.126	0	0	0	L	L	0	L
MLM-FE	-0.122	0	L	L	L	L	0	L
MLM-RE	-0.132	0	0	0	L	L	0	L

method	bias.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
MLM-RE	-0.127	S	0	L	S	S	0	L
MLM-RE	-0.117	0	L	0	S	L	L	L
MLM-RE	-0.113	0	0	0	L	L	0	L
MLM-RE	-0.11	S	L	0	L	0	0	L

What does the error look like for each estimator

Main Bullet Points:

*For the person weighted estimators, the unbiased estimator both has less error and is more stable than the biased estimator.

*For the cluster weighted estimators, the unbiased estimator generally has less error than the biased estimators but more instablity.

*The MLM estimators are the most stable, which fits with theory.

*High RMSE and estimate instability is driven by a low number of clusters, the interaction of a low number of clusters and high cluster level ICC, and high cluster level treatment effect variance.

*Re the question on pg. 39, looking at the second boxplot shows that true S.E. do vary across estimators with the true standard error in a given DGP often 40% bigger for some estimates than the median estimate in that DPG.

Description of Analysis:

*The first graph is a boxplot of the normalized RMSE across the different scenarios by method. The normalized RMSE is the RMSE for given method divided by the median RMSE across methods within a given simulation scenario. Therefore each data point is adjusted for the underlying base RMSE from a given set of DGPs and represents the RMSE driven by the estimation method.

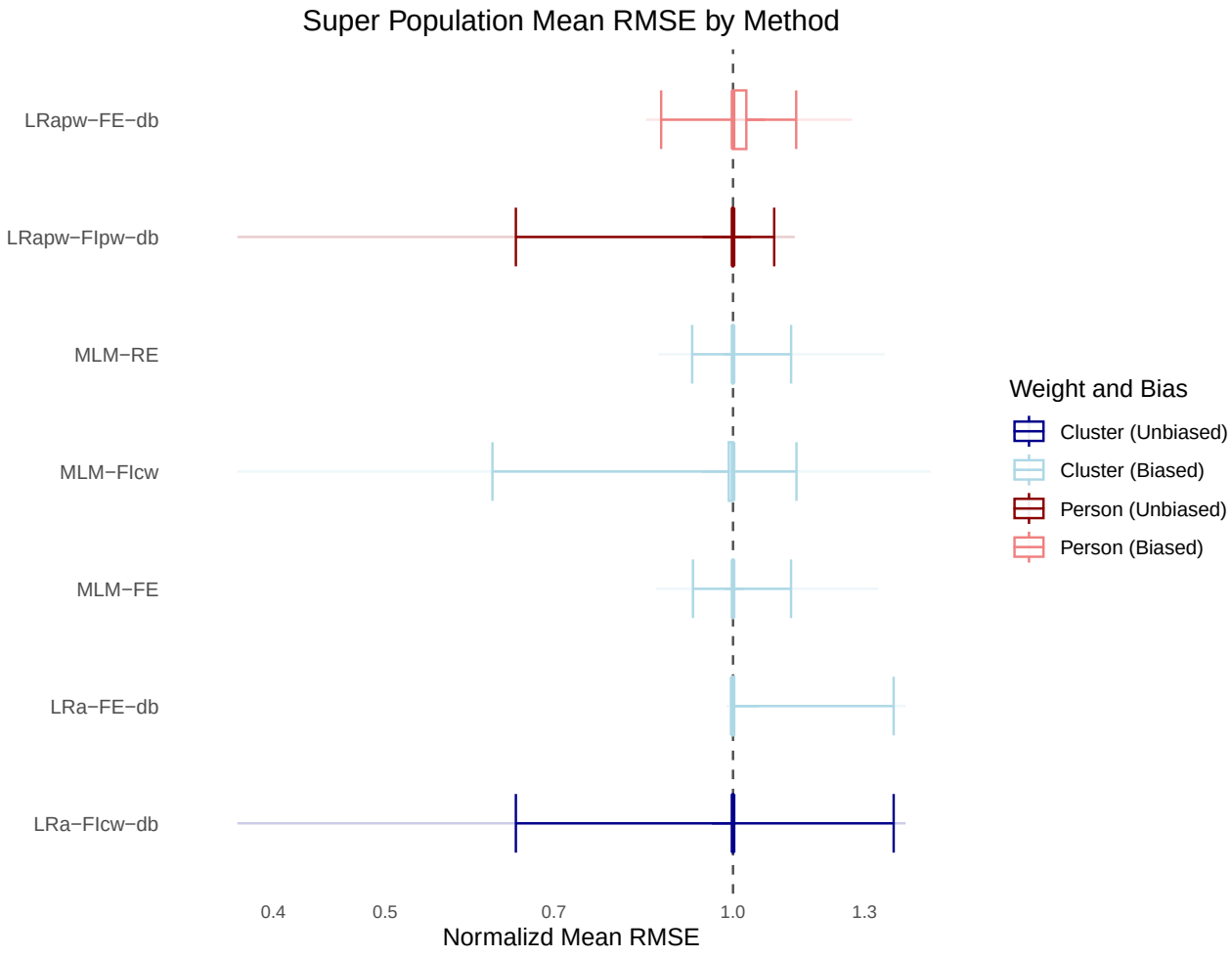
*The second graph is a boxplot of the normalized estimate standard deviation across the different scenarios by method. This is also normalized by dividing by the median estimate standard deviation within a given simulation scenario.

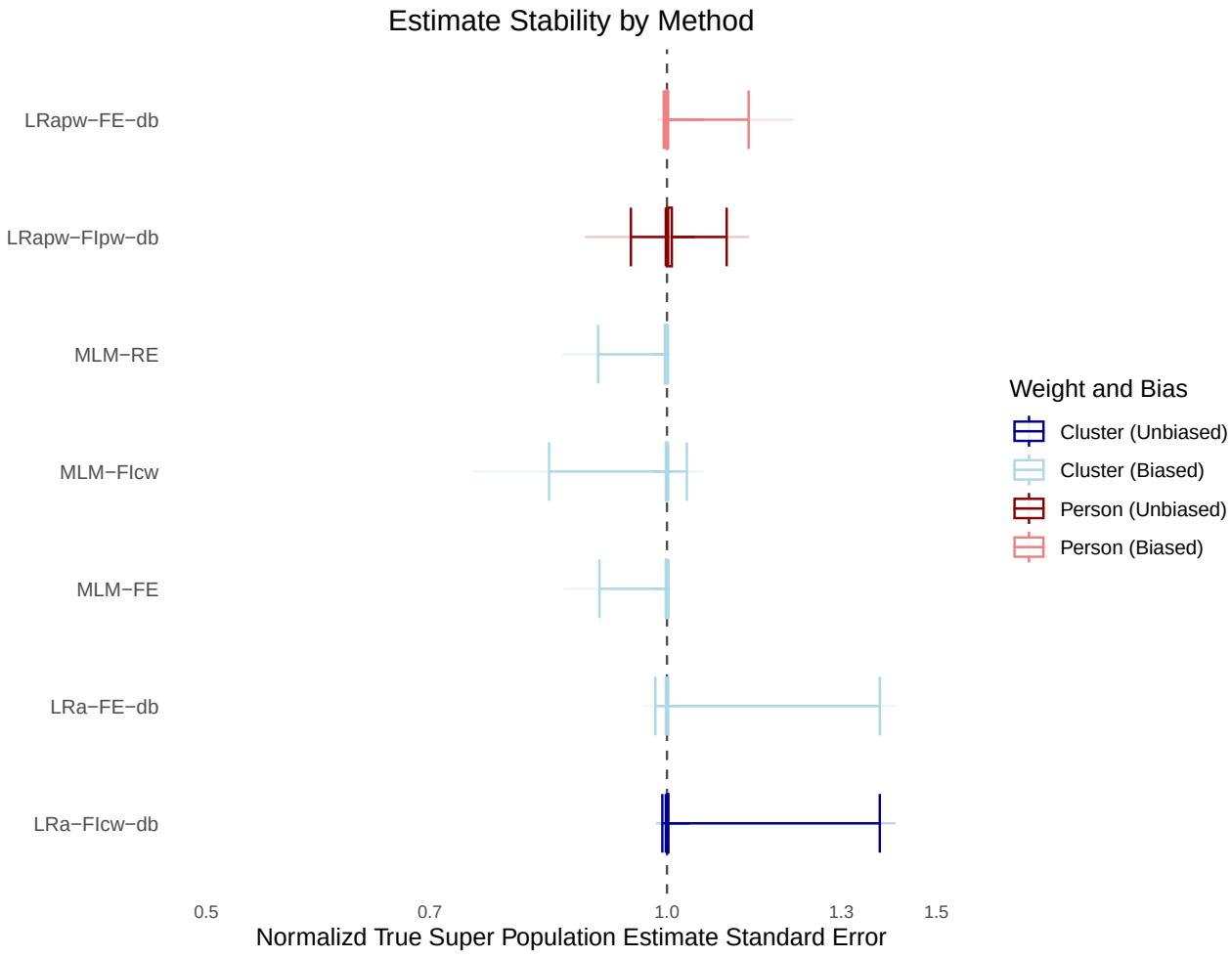
*The first table contains the regression coefficients from a regression that predicts the RMSE using different data generating parameters and the second table is from regressions predicting the estimate standard deviation. Each regression is run separately by estimator and the RMSE is against the targeted estimator.

*The second table contains the DGPs that produce the normalized RMSE (against the targeted estimator) for each method, conditional on the magnitude of the being at least 1.10.

*The third table contains the regression coefficients from a regression that predicts the true superpopulation estimate standard error using different data generating parameters.

*The fourth table contains the five DGPs that produce the most normalized super population estimator standard error for each method, conditional on the magnitude of the being at least 1.10.


Super Population Mean RMSE by Method
LRapw−FE−db
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRa−FE−db
LRa−FIcw−db
0.4
0.5
0.7
1.0
1.3
Normalizd Mean RMSE
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)



The Effect of Different Data Generating Parameters on the RMSE

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	.071***	.067***	.074***	.070***	.072***	.069***	.072***
block_typeS	.002	.002	.002	.002	.002	.002	.002
block_typeL	.015***	<.001	.014***	<.001	.017***	.001	.015***
J.barS	-.013***	-.011***	-.012***	-.010**	-.013***	-.011***	-.013***
J.barL	-.024***	-.024***	-.022***	-.022***	-.025***	-.025***	-.024***
size.impact.cor.schoolL	-.005.	-.003	-.005	-.003	-.006*	-.005.	-.006.
sd.schoolL	.024***	.026***	.025***	.026***	.024***	.024***	.023***
ICC.schoolL	.090***	.094***	.087***	.090***	.092***	.097***	.093***
variable.nTRUE	.003	.005.	-.005	-.003	-.002	-.001	-.002
cv.schoolS	-.002	.001	-.004	-.001	-.003	<.001	-.003
cv.schoolL	.003	.004	-.007.	-.005.	-.003	-.004	-.003
num.cov.schoolS	-.004	-.001	-.004	-.001	-.004	-.001	-.003
num.cov.schoolL	.001	<.001	.001	<.001	.001	<.001	.001
block_typeS:J.barS	-.002	-.002	-.002	-.002	-.002	-.002	-.002
block_typeL:J.barS	.003	.004.	.002	.003	.002	.003	.003
block_typeS:J.barL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeL:J.barL	.003	.003	.001	.001	.002	.002	.003
block_typeS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: size.impact.cor.schoolL	<.001	.002	<.001	.003	-.002	.002	-.001
block_typeS:sd.schoolL	.001	.001	.001	.001	.001	.001	.001
block_typeL:sd.schoolL	<.001	.002	-.001	.002	-.002	.002	-.002
block_typeS:ICC.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:ICC.schoolL	.001	.009***	<.001	.008***	.001	.009***	.001
block_typeS: variable.nTRUE	.001	.001	.001	.001	.001	.001	.001

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
block_typeL: variable.nTRUE	.002	.002	.008***	.005**	.002	.002	.002
block_typeS:cv.schoolS	-.001	-.001	<.001	<.001	<.001	<.001	<.001
block_typeL:cv.schoolS	.002	.002	.004	.004.	.002	.002	.002
block_typeS:cv.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeL:cv.schoolL	.004	.004.	.011***	.008***	.005.	.004.	.004.
block_typeS: num.cov.schoolS	.001	.001	.001	.001	.001	.001	.001
block_typeL: num.cov.schoolS	-.003	-.001	-.003	-.001	-.003	-.001	-.003
block_typeS: num.cov.schoolL	.001	.001	.001	.001	.001	.001	.001
block_typeL: num.cov.schoolL	-.002	-.002	-.002	-.002	-.002	-.002	-.002
J.barS: size.impact.cor.schoolL	.001	<.001	.001	<.001	.001	<.001	.001
J.barL: size.impact.cor.schoolL	-.002	-.003	-.002	-.003	-.001	-.002	-.001
J.barS:sd.schoolL	-.002	-.002	-.002	-.002	-.001	-.002	-.001
J.barL:sd.schoolL	-.003	-.003.	-.003	-.003.	-.002	-.003	-.002
J.barS:ICC.schoolL	-.013***	-.015***	-.014***	-.016***	-.014***	-.016***	-.014***
J.barL:ICC.schoolL	-.028***	-.028***	-.029***	-.030***	-.029***	-.030***	-.030***
J.barS:variable.nTRUE	.001	<.001	.001	<.001	.002	.001	.001
J.barL:variable.nTRUE	-.003	-.004.	-.004.	-.004*	-.001	-.002	-.001
J.barS:cv.schoolS	.001	<.001	.001	<.001	.001	<.001	.001
J.barL:cv.schoolS	<.001	<.001	<.001	<.001	<.001	.001	<.001
J.barS:cv.schoolL	-.003	-.003	-.004	-.003	-.002	-.002	-.002
J.barL:cv.schoolL	-.004.	-.003	-.005.	-.003	-.002	-.001	-.001
J.barS:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:num.cov.schoolS	.002	.002	.002	.003	.002	.002	.002
J.barS:num.cov.schoolL	-.002	-.001	-.002	-.001	-.002	-.001	-.002
J.barL:num.cov.schoolL	-.003	-.001	-.003	-.001	-.004	-.002	-.004
size.impact.cor.schoolL: sd.schoolL	<.001	-.001	<.001	-.001	.002	.002	.002
size.impact.cor.schoolL: ICC.schoolL	.003*	.002	.004*	.003.	.002	<.001	.002
size.impact.cor.schoolL: variable.nTRUE	.003	.003.	.003	.003.	.004*	.006***	.005**
size.impact.cor.schoolL: cv.schoolS	.002	<.001	.002	<.001	.002	.001	.002
size.impact.cor.schoolL: cv.schoolL	.002	.001	.001	.001	.004.	.004*	.003.
size.impact.cor.schoolL: num.cov.schoolS	.002	.002	.002	.002	.002	.002	.002
size.impact.cor.schoolL: num.cov.schoolL	.001	.002	.001	.002	.001	.002	.001
sd.schoolL:ICC.schoolL	-.009***	-.010***	-.010***	-.011***	-.012***	-.014***	-.012***
sd.schoolL:variable.nTRUE	-.002	-.003.	-.001	-.001	.001	.001	.001
sd.schoolL:cv.schoolS	-.004*	-.004*	-.004.	-.004.	-.004.	-.003	-.003
sd.schoolL:cv.schoolL	-.004.	-.003	-.001	-.001	<.001	.002	.001
sd.schoolL: num.cov.schoolS	<.001	<.001	<.001	-.001	<.001	-.001	<.001
sd.schoolL: num.cov.schoolL	-.002	-.001	-.002	-.001	-.001	-.001	-.001
ICC.schoolL: variable.nTRUE	-.001	-.003.	.009***	.008***	.002	<.001	.001
ICC.schoolL:cv.schoolS	.001	-.001	.004.	.002	.001	<.001	.001
ICC.schoolL:cv.schoolL	-.005*	-.006**	.008***	.007***	-.001	-.002	-.001
ICC.schoolL: num.cov.schoolS	-.001	-.002	-.001	-.002	-.001	-.002	-.001
ICC.schoolL: num.cov.schoolL	-.001	-.001	-.001	-.002	-.001	-.001	-.001
variable.nTRUE:cv.schoolS	.002	.001	.004*	.003.	.001	.001	.001
variable.nTRUE:cv.schoolL	.015***	.015***	.016***	.014***	.008***	.008***	.008***
variable.nTRUE: num.cov.schoolS	-.004.	-.004*	-.004.	-.004*	-.004.	-.004*	-.004.
variable.nTRUE: num.cov.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
cv.schoolS: num.cov.schoolS	.007**	.005*	.007**	.005*	.007**	.005*	.006*
cv.schoolL: num.cov.schoolS	.006*	.005*	.006*	.005*	.006*	.005*	.006*
cv.schoolS: num.cov.schoolL	.002	.001	.003	.001	.002	.001	.002
cv.schoolL: num.cov.schoolL	.003	.002	.003	.002	.004	.002	.003

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Largest Normalized RMSE by Method

method	RMSE.s.estimand.norm	RMSE.sestimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	1.41	0.0635	L	0	0	L	S	0	0
LRa-FE-db	1.4	0.0632	L	0	0	L	0	L	0
LRa-FE-db	1.4	0.0635	L	0	0	L	S	L	0
LRa-FE-db	1.4	0.0633	L	0	0	L	L	L	0
LRa-FE-db	1.4	0.0811	S	0	0	L	0	L	0
LRapw-FE-db	1.27	0.098	L	0	0	L	S	0	L
LRapw-FE-db	1.25	0.113	L	L	0	S	S	L	L
LRapw-FE-db	1.17	0.202	L	L	L	L	0	L	L
LRapw-FE-db	1.16	0.247	0	L	L	L	L	0	L
LRapw-FE-db	1.16	0.142	0	0	0	L	S	0	L
MLM-FE	1.34	0.0638	L	L	0	S	0	L	L
MLM-FE	1.32	0.0996	L	L	0	L	L	L	0
MLM-FE	1.31	0.0888	L	L	0	L	S	L	L
MLM-FE	1.31	0.0941	L	L	0	L	0	L	L
MLM-FE	1.31	0.1	L	L	0	L	S	L	0
MLM-FIcw	1.48	0.1	L	L	0	L	S	L	L
MLM-FIcw	1.39	0.0999	L	L	0	L	0	L	L
MLM-FIcw	1.33	0.101	L	L	0	L	L	L	0
MLM-FIcw	1.32	0.101	L	L	0	L	S	L	0
MLM-FIcw	1.32	0.101	L	L	0	L	0	L	0
MLM-RE	1.35	0.0646	L	L	0	S	0	L	L
MLM-RE	1.34	0.0963	L	L	0	L	0	L	L
MLM-RE	1.32	0.0996	L	L	0	L	L	L	0
MLM-RE	1.32	0.0891	L	L	0	L	S	L	L
MLM-RE	1.31	0.1	L	L	0	L	S	L	0
LRa-FIcw-db	1.41	0.0635	L	0	0	L	S	0	0

method	RMSE.s.estimand.norm	RMSE.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FIcw-db	1.4	0.0632	L	0	0	L	0	L	0
LRa-FIcw-db	1.4	0.0635	L	0	0	L	S	L	0
LRa-FIcw-db	1.4	0.0633	L	0	0	L	L	L	0
LRa-FIcw-db	1.4	0.0811	S	0	0	L	0	L	0
LRapw-FIpw-db	1.13	0.112	L	0	L	L	L	0	L
LRapw-FIpw-db	1.12	0.177	L	0	L	L	S	0	L
LRapw-FIpw-db	1.11	0.238	S	L	L	L	L	L	L

The Effect of Different Data Generating Parameters on the Estimate Instability

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	.067***	.067***	.069***	.070***	.068***	.069***	.068***
block_typeS	.002	.002	.002	.002	.002	.002	.002
block_typeL	<.001	<.001	-.001	<.001	.001	<.001	.001
J.barS	-.011***	-.011***	-.010**	-.010**	-.011***	-.011***	-.011***
J.barL	-.024***	-.024***	-.022***	-.022***	-.024***	-.024***	-.023***
size.impact.cor.schoolL	-.003	-.003	-.002	-.003	-.002	-.003	-.002
sd.schoolL	.026***	.026***	.027***	.026***	.026***	.026***	.026***
ICC.schoolL	.094***	.094***	.091***	.090***	.093***	.092***	.093***
variable.nTRUE	.005.	.005.	-.003	-.003	.001	.001	.001
cv.schoolS	.001	.001	-.001	-.001	<.001	<.001	<.001
cv.schoolL	.004	.004	-.006.	-.005	-.001	-.001	-.001
num.cov.schoolS	-.001	-.001	-.001	-.001	-.001	-.001	-.001
num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeS:J.barS	-.002	-.002	-.002	-.002	-.002	-.002	-.002
block_typeL:J.barS	.004	.004.	.003	.003	.003	.004	.003
block_typeS:J.barL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeL:J.barL	.003	.003	.001	.002	.002	.002	.002
block_typeS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: size.impact.cor.schoolL	.002	.002	.002	.003	.002	.002	.002
block_typeS:sd.schoolL	.001	.001	.001	.001	.001	.001	.001
block_typeL:sd.schoolL	.003	.002	.001	.001	.002	.002	.002
block_typeS:ICC.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:ICC.schoolL	.008***	.009***	.006***	.008***	.007***	.009***	.006***
block_typeS: variable.nTRUE	.001	.001	.001	.001	.001	.001	.001
block_typeL: variable.nTRUE	.002	.002	.008***	.005**	.003.	.002	.003.
block_typeS:cv.schoolS	-.001	-.001	<.001	<.001	-.001	-.001	<.001
block_typeL:cv.schoolS	.002	.002	.004.	.004.	.002	.002	.002
block_typeS:cv.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeL:cv.schoolL	.004.	.004.	.011***	.008***	.006**	.004.	.006**
block_typeS: num.cov.schoolS	.001	.001	.001	.001	.001	.001	.001

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
block_typeL: num.cov.schoolS	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeS: num.cov.schoolL	.001	.001	.001	.001	.001	.001	.001
block_typeL: num.cov.schoolL	-.002	-.002	-.002	-.002	-.002	-.002	-.002
J.barS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL: size.impact.cor.schoolL	-.003	-.003	-.003	-.003	-.003	-.003	-.003
J.barS:sd.schoolL	-.002	-.002	-.002	-.002	-.002	-.002	-.002
J.barL:sd.schoolL	-.003.	-.003.	-.003.	-.003.	-.003.	-.004.	-.003.
J.barS:ICC.schoolL	-.015***	-.015***	-.016***	-.016***	-.015***	-.015***	-.015***
J.barL:ICC.schoolL	-.029***	-.028***	-.030***	-.030***	-.029***	-.029***	-.029***
J.barS:variable.nTRUE	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:variable.nTRUE	-.004*	-.004.	-.005*	-.004*	-.003	-.003	-.003
J.barS:cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barS:cv.schoolL	-.003	-.003	-.004	-.003	-.003	-.003	-.003
J.barL:cv.schoolL	-.003	-.003	-.004.	-.003	-.002	-.002	-.002
J.barS:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:num.cov.schoolS	.002	.002	.002	.003	.002	.002	.002
J.barS:num.cov.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
J.barL:num.cov.schoolL	-.001	-.001	-.002	-.001	-.001	-.001	-.001
size.impact.cor.schoolL: sd.schoolL	-.001	-.001	-.002	-.001	-.001	-.001	-.001
size.impact.cor.schoolL: ICC.schoolL	.002	.002	.003.	.003.	.002	.002	.002
size.impact.cor.schoolL: variable.nTRUE	.003.	.003.	.002	.003.	.003.	.003.	.003.
size.impact.cor.schoolL: cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
size.impact.cor.schoolL: cv.schoolL	.001	.001	.001	.001	.001	.001	.001
size.impact.cor.schoolL: num.cov.schoolS	.002	.002	.002	.002	.002	.002	.002
size.impact.cor.schoolL: num.cov.schoolL	.002	.002	.002	.002	.002	.002	.002
sd.schoolL:ICC.schoolL	-.010***	-.010***	-.011***	-.011***	-.011***	-.011***	-.011***
sd.schoolL:variable.nTRUE	-.003.	-.003.	-.002	-.001	-.002	-.001	-.002
sd.schoolL:cv.schoolS	-.004*	-.004*	-.004*	-.004.	-.004*	-.004*	-.004*
sd.schoolL:cv.schoolL	-.003.	-.003	-.001	-.001	-.002	-.001	-.002
sd.schoolL: num.cov.schoolS	<.001	<.001	-.001	-.001	-.001	-.001	-.001
sd.schoolL: num.cov.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
ICC.schoolL: variable.nTRUE	-.003.	-.003.	.008***	.008***	.002	.003.	.002
ICC.schoolL:cv.schoolS	-.001	-.001	.002	.002	<.001	<.001	<.001
ICC.schoolL:cv.schoolL	-.006**	-.006**	.007***	.007***	<.001	.001	<.001
ICC.schoolL: num.cov.schoolS	-.002	-.002	-.002	-.002	-.002	-.002	-.002
ICC.schoolL: num.cov.schoolL	-.001	-.001	-.002	-.002	-.001	-.001	-.001
variable.nTRUE:cv.schoolS	.001	.001	.004*	.003.	<.001	<.001	<.001
variable.nTRUE:cv.schoolL	.015***	.015***	.016***	.014***	.005**	.004*	.005**
variable.nTRUE: num.cov.schoolS	-.004*	-.004*	-.004*	-.004*	-.004*	-.004*	-.004*
variable.nTRUE: num.cov.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
cv.schoolS: num.cov.schoolS	.004.	.005*	.004.	.005*	.004.	.005*	.004*
cv.schoolL: num.cov.schoolS	.005*	.005*	.005*	.005*	.005*	.005*	.005*
cv.schoolS: num.cov.schoolL	.001	.001	.001	.001	.001	.001	.001

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
cv.schoolL: num.cov.schoolL	.002	.002	.002	.002	.002	.002	.002

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Largest Normalized Estimate Standard Error by Method

method	SE s.norm	SE.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	1.41	0.0635	L	0	0	L	S	0	0
LRa-FE-db	1.4	0.0632	L	0	0	L	0	L	0
LRa-FE-db	1.4	0.0635	L	0	0	L	S	L	0
LRa-FE-db	1.4	0.0633	L	0	0	L	L	L	0
LRa-FE-db	1.4	0.0811	S	0	0	L	0	L	0
LRapw-FE-db	1.21	0.0918	L	0	0	L	S	0	L
LRapw-FE-db	1.19	0.139	0	0	0	L	L	0	L
LRapw-FE-db	1.19	0.0731	S	0	0	S	L	0	L
LRapw-FE-db	1.17	0.184	L	L	L	L	0	L	L
LRapw-FE-db	1.15	0.222	0	0	L	L	L	0	L
LRa-FIcw-db	1.41	0.0635	L	0	0	L	S	0	0
LRa-FIcw-db	1.4	0.0632	L	0	0	L	0	L	0
LRa-FIcw-db	1.4	0.0635	L	0	0	L	S	L	0
LRa-FIcw-db	1.4	0.0633	L	0	0	L	L	L	0
LRa-FIcw-db	1.4	0.0811	S	0	0	L	0	L	0
LRapw-FIpw-db	1.13	0.112	L	0	L	L	L	0	L
LRapw-FIpw-db	1.12	0.134	L	0	L	L	0	L	L
LRapw-FIpw-db	1.12	0.177	L	0	L	L	S	0	L
LRapw-FIpw-db	1.11	0.213	0	0	L	L	L	0	L
LRapw-FIpw-db	1.11	0.123	L	0	L	L	L	L	L

What do the estimated standard errors look like for each estimator

Main Bullet Points:

*This section answers the rest of the questions from pg.39 (i.e. are certain SE estimators more accurate than others)

*For the cluster targeted estimates the unbiased estimators produce more accurate standard errors than the biased estimators (graph one). The unbiased estimators produce more unstable estimates than the biased estimators, but in 90% of the DGP we looked at they aren't a lot more unstable (graph four).

*For the person targeted estimates the unbiased estimators produce more accurate standard errors than the biased estimators (graph one), except when the school ICC is zero and the blocks are all very dissimilar from each other and their is high variance in the cluster sizes. if all three of these risk factors are not present than the unbiased estimator standard estimates are generally fine (graph two). The unbiased estimators produce more unstable estimates than the biased estimators unless all three of these factors are present (graphs four and five). However they aren't a lot more unstable, 90% of the time they're within about 10% of the median instability.

*This section also answers the question from pg. 38 about the instability of the estimated standard erros for the fully interacted models, what I've been calling hte unbiased estimators. The simulations do not really show that the standard errors from the fully interacted models are lot more unstable than the other standard errors. We should talk about this more, and what would be helpful for me to dig into more.

Description of Analysis:

*The first graph is a boxplot of the estimated standard error accuracy across the different scenarios by method. Standard error accuracy is quantified as the estimated standard error over the true super population standard error. Estimated standard errors are most accurate if the number is close to one. Numbers below one represent under estimated standard errors and numbers above one represent over estimated standard errors.

*The second graph is the same as the first but for just the person weighted fully interacted estimators broken out by identified "risk factors" for poorly estimated standard errors. In the graph on the left the school level ICC is zero, the blocks are all messy, and there is high variance in the cluster size and the graph on the right does not have all three of these things at the same time.

*The third graph is a boxplot of the standard deviation of the estimated standard error across the different replications by method.

*The forth graph is a boxplot of the normalized standard deviation of the estimated standard error across the different replications by method.

*The fifth graph is the same as the fourth graph but broken by the same risk factors as before.

- I was having trouble conceptualizing the scale of standard error inflation, so for the regression I converted it to percentage inaccuracy. The regression's for table one are therefore predicting the magnitude of ((Mean Estimated SE - True Super Population SE) / True Super Population SE) * 100.

*The second table contains the five DGPs that produce the most inaccurate standard error estimates for each method, conditional on the magnitude of the inaccuracy being at least .1.

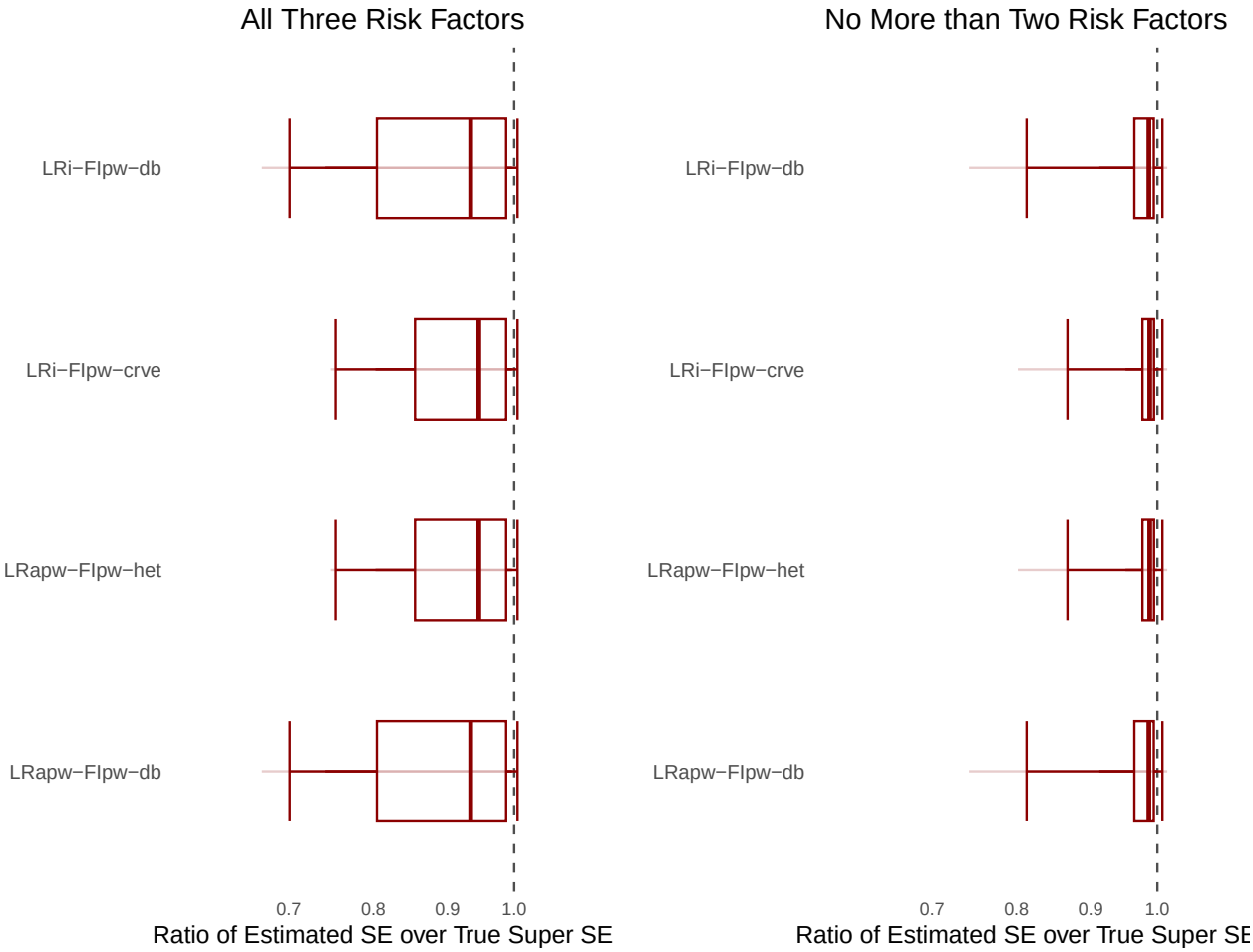
*The third table contains the regression coefficients from a regression that predicts the standard deviation of the standard errors across simulations using different data generating parameters.

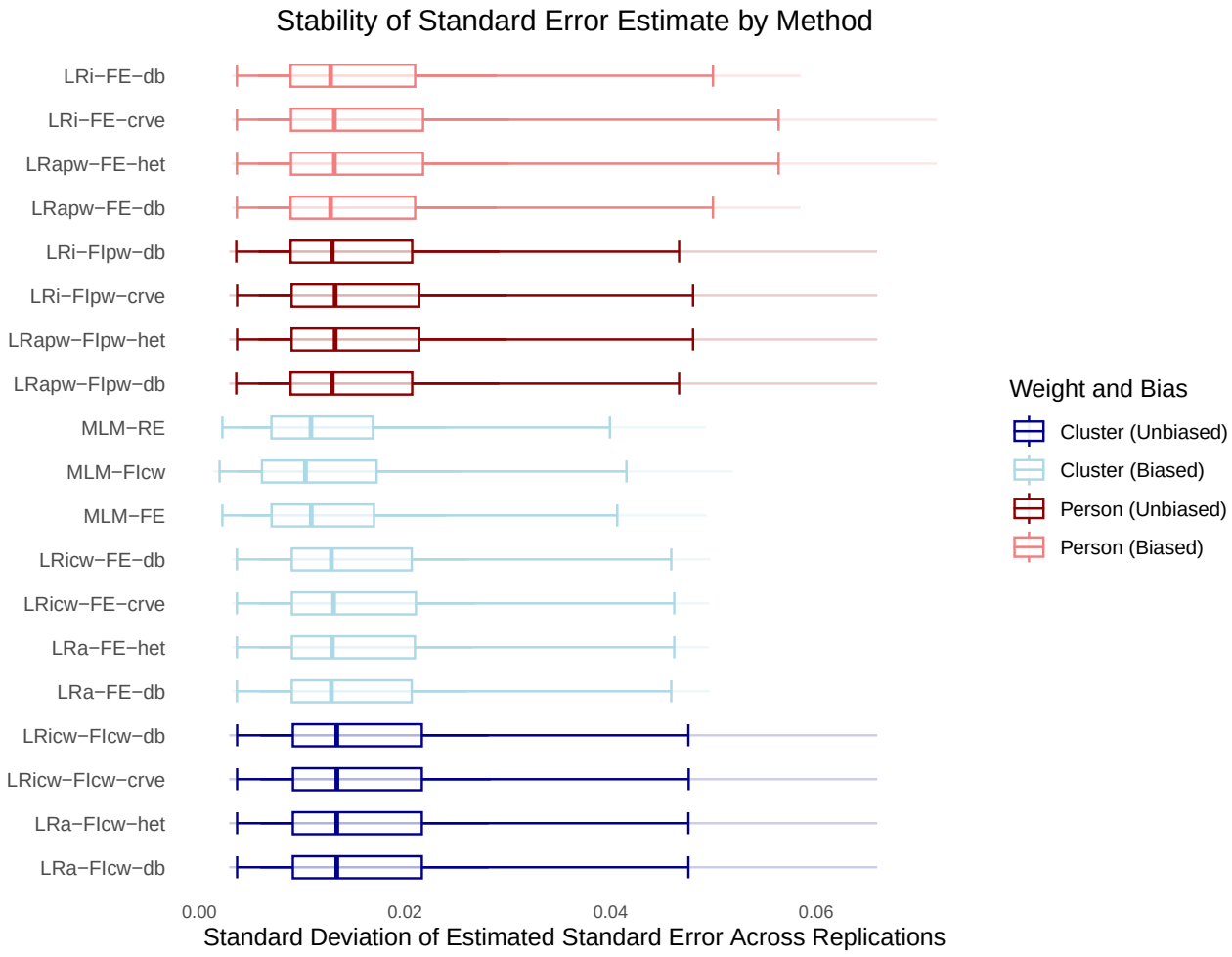
*The fourth table contains the five DGPs that produce the largest standard deviation in the standard error estimates for each method, conditional on the magnitude being at least .05.

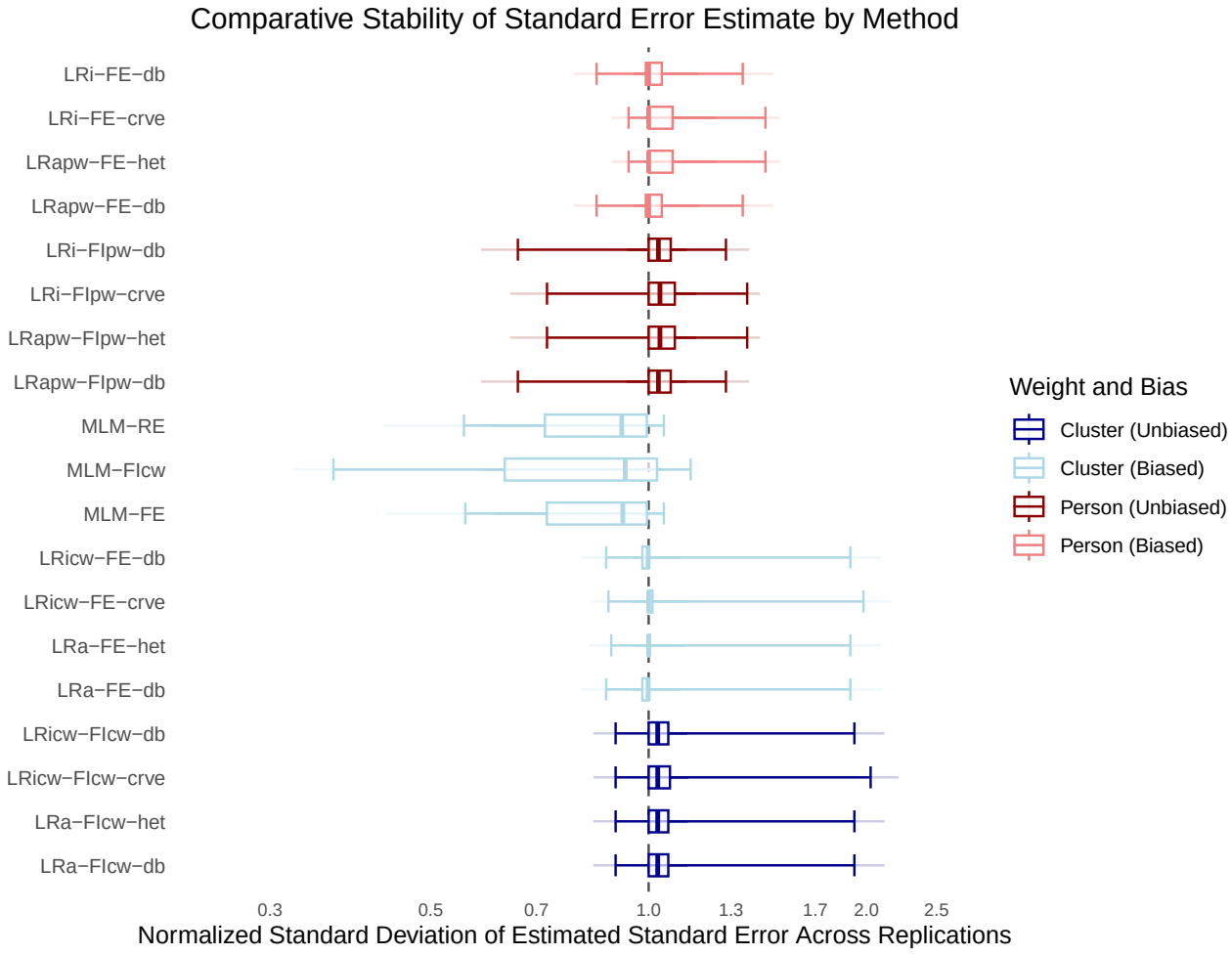
*The fifth table contains the five DGPs that produce the largest normalized standard deviation in the standard error estimates for each method, conditional on the magnitude being at least 1.10.


Super Population Estimated Standard Error Accuracy by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.7
1.0
1.3
1.7
2.0
2.5
Ratio of Estimated SE over True Super SE
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)

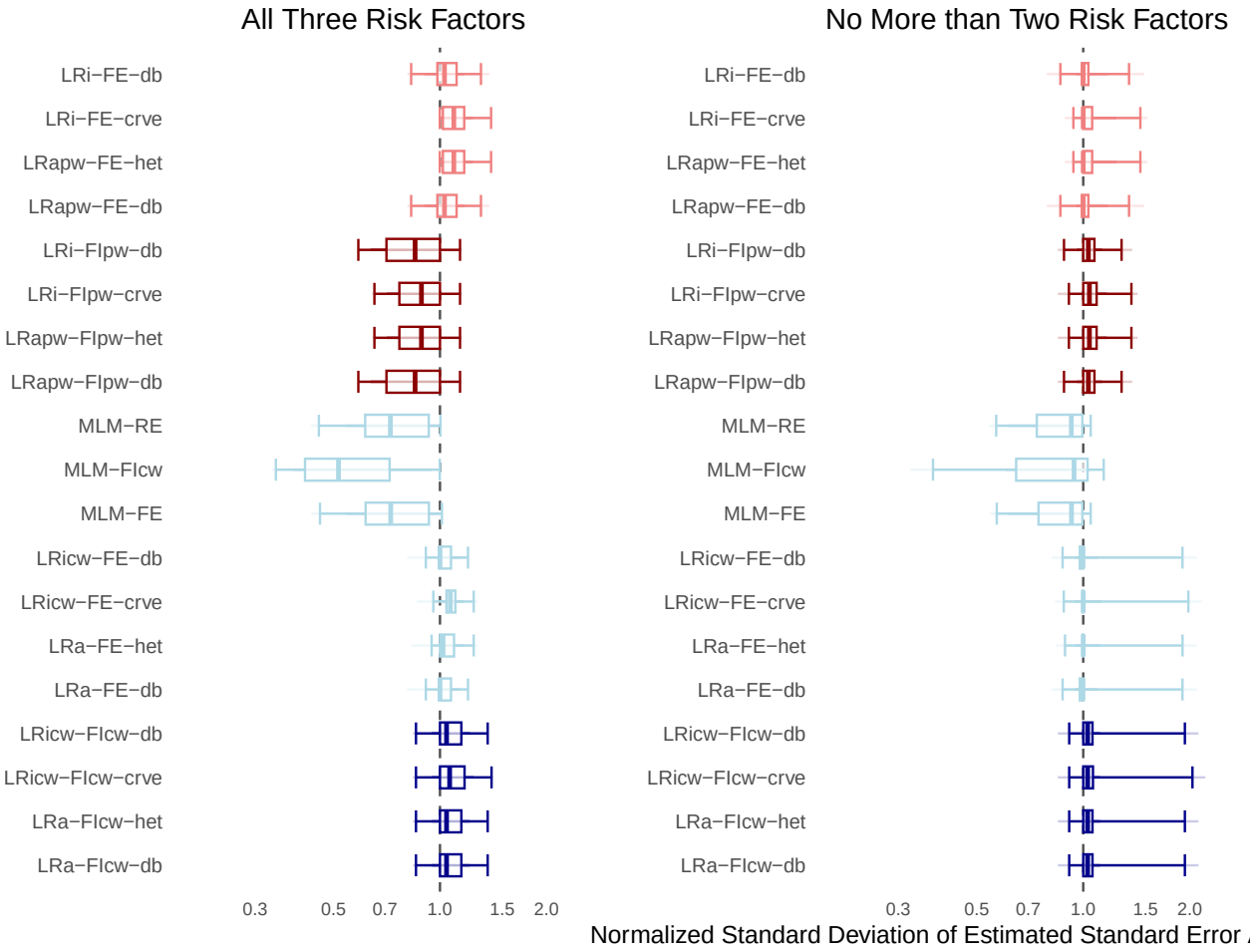
The Risk Factors are School Level ICC = 0 & Blocks All Different & High Variance in Cluster Size




Stability of Standard Error Estimate by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.00
0.02
0.04
0.06
Standard Deviation of Estimated Standard Error Across Replications
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)


Comparative Stability of Standard Error Estimate by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.3
0.5
0.7
1.0
1.3
1.7
2.0
2.5
Normalized Standard Deviation of Estimated Standard Error Across Replications
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)

The Risk Factors are School Level ICC = 0 & Blocks All Different & High Variance in Cluster Size



The Effect of Different Data Generating Parameters on % Points Estimated SE Off from True Super SE

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	5.842**	6.070**	1.069***	5.501**	5.975**	.990*	1.092**	5.679**	1.217***	8.017***	5.468***	8.038***
block_typeS	.785	.885	.288.	.794	1.200	.146	.145	.947	.313.	.157	-.929.	.170
block_typeL	79.662***	82.928***	.755***	81.702***	84.100***	-.307	-.166	84.159***	.977***	57.767***	1.269*	57.528***
J.barS	-.501	-.572	-.410*	-.366	-.595	.243	.145	-.525	-.495**	-1.450	-	-1.429
J.barL	-.546	-.854	-.676***	-.829	-.887	.908*	.536	-.905	-.780***	-3.217*	1.872*** -	-3.246*
size.impact.cor.schoolL	-2.549	-2.439	-.060	-3.061.	-2.992*	-.578	-.517	-2.456	.009	-1.619	3.597*** -.292	-1.614
sd.schoolL	-5.786***	-5.976***	-.041	-6.463***	-6.511***	.074	.182	-5.977***	-.071	-4.594***	.238	-4.654***
ICC.schoolL	-9.989***	- 10.379***	-.105	-9.942***	-9.782***	.849*	.908**	-10.238***	-.391**	- 10.638***	- 4.002***	- 10.585***
variable.nTRU E	-.076	-.245	.109	1.458	.164	-.769*	-1.147***	.289	-.309*	1.030	.138	1.063
cv.schoolS	.132	-.093	.211	.784	.259	-.079	-.144	.031	.144	.733	.020	.639
cv.schoolL	-.230	-.296	.599***	1.418	-.034	-.427	-.929*	.381	.040	-.510	-.315	-.555
num.cov.schoo lS	-1.140	-1.279	.017	-1.376	-1.657	-.011	-.005	-1.286	-.001	-.418	.024	-.432
num.cov.schoo lL	.135	.174	.013	-.072	-.183	-.027	-.085	.250	.082	-.201	-.084	-.234
block_typeS: J.barS	.250	.183	-.103	.153	.079	-.169	-.196	.138	-.074	.985	.919*	.978
block_typeL: J.barS	-1.105	-1.001	-.107	-.396	-.998	.089	-.178	-.756	-.059	-1.544	1.624***	-1.557
block_typeS: J.barL	.422	.361	-.096	.338	.271	-.102	-.199	.383	.039	.967	.993*	.959
block_typeL: J.barL	-2.036	-2.360	-.462***	.170	-1.841	-.569.	-.907**	-1.648	-.217.	-3.667***	1.827***	-3.489**
block_typeS: size.impact.cor.schoolL	.021	.025	-.022	.021	-.020	.023	.028	.035	-.044	.069	.113	.076
block_typeL: size.impact.cor.schoolL	-2.313.	-2.316*	.126	-2.014.	-1.956.	.408	.378	-2.270.	.077	-.906	-.793*	-.822
block_typeS: sd.schoolL	-.281	-.337	.016	-.090	-.614	.062	.052	-.309	.016	-1.299	-.543.	-1.294
block_typeL: sd.schoolL	- 17.243***	- 17.420***	-.024	- 16.305***	-16.905***	-.988***	-.910***	-17.414***	-.059	- 10.763***	3.666***	- 10.700***
block_typeS: ICC.schoolL	-.614	-.633	.002	-.344	-.851	-.200	-.135	-.645	-.028	-.271	.646*	-.266

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
block_typeL: ICC.schoolL	- 47.760***	- 49.338***	-.140	- 47.155***	-49.943***	-1.918***	-1.798***	-49.892***	-.370***	- 34.502***	- 2.556***	- 34.672***
block_typeS: variable.nTRUE	-.383	-.383	-.036	-.701	-.299	.357	.350	-.471	-.110	-.143	.016	-.148
block_typeL: variable.nTRUE	-5.922***	-6.151***	.138	- 13.397***	-8.641***	4.563***	4.528***	-8.710***	-.217*	-3.898***	-.235	-3.815***
block_typeS: cv.schoolS	-.171	-.165	-.191	-.501	-.250	-.010	-.033	-.224	-.217.	.018	-.156	-.003
block_typeL: cv.schoolS	-.005	-.016	.043	-4.063**	-2.210.	2.014***	1.957***	-.585	-.024	.123	-.448	.114
block_typeS: cv.schoolL	-.449	-.458	-.217.	-.596	-.248	.192	.205	-.532	-.302*	.185	.238	.195
block_typeL: cv.schoolL	-6.736***	-6.960***	.067	- 13.890***	-8.502***	4.734***	4.738***	-10.231***	-.398**	-3.676***	-.357	-3.635**
block_typeS: num.cov.schoolS	-.149	-.146	-.047	-.102	-.139	.109	.101	-.154	-.062	-.146	-.173	-.153
block_typeL: num.cov.schoolS	-1.002	-1.007	-.025	-1.112	-1.155	-.075	-.051	-.997	.028	-.187	.039	-.127
block_typeS: num.cov.schoolL	.033	.032	.071	-.074	-.044	-.008	.046	-.029	.037	.020	-.003	.017
block_typeL: num.cov.schoolL	.687	.563	.141	.211	.003	.339	.298	.495	.132	-1.117	-.615	-1.080
J.barS: size.impact.cor.schoolL	.592	.661	.078	.571	.656	.152	.113	.682	.095	.243	.324	.212
J.barL: size.impact.cor.schoolL	<.001	.117	.011	.208	.217	-.110	-.094	.177	.052	-.364	.404	-.369
J.barS: sd.schoolL	.224	.273	.131	.024	.294	.157	.116	.212	.150	-1.172	-.547.	-1.243
J.barL: sd.schoolL	.921	1.003	.100	.563	.915	.221	.130	.899	.129	.266	.031	.225
J.barS: ICC.schoolL	.264	.349	.126	-.003	.202	-.180	-.263	.213	.186.	1.470	.992**	1.464
J.barL: ICC.schoolL	1.069	1.189	.282**	.304	.638	-.311	-.380.	.932	.258*	2.851**	1.347***	2.827**
J.barS: variable.nTRUE	1.003	1.049	.057	.728	.945	-1.187***	-.666**	.982	.132	-.344	-.035	-.342
J.barL: variable.nTRUE	-.825	-.623	-.046	-.844	-.530	-2.689***	-1.491***	-.728	-.139	-.577	.729*	-.588
J.barS: cv.schoolS	-.735	-.514	-.003	-1.145	-.949	-.490	-.289	-.542	-.018	-.905	-.113	-.883
J.barL: cv.schoolS	.316	.432	.003	.199	.280	-1.051***	-.578*	.394	-.112	.937	.413	1.056
J.barS: cv.schoolL	-1.044	-1.113	-.328**	-1.047	-.834	-1.707***	-1.126***	-1.186	-.200	-.012	.246	-.008
J.barL: cv.schoolL	-.871	-.903	-.366**	-.782	-.613	-3.277***	-1.953***	-1.023	-.390**	.435	.183	.508
J.barS: num.cov.schoolS	-.088	-.044	-.022	.118	.303	-.033	-.006	-.020	-.057	.245	-.264	.257
J.barL: num.cov.schoolS	.702	.866	.052	.515	.826	.310	.254	.840	.052	.623	.345	.611
J.barS: num.cov.schoolL	-.500	-.606	-.219.	-.579	-.430	-.122	-.100	-.605	-.259*	.635	-.290	.711
J.barL: num.cov.schoolL	-1.613	-1.466	-.122	-1.298	-1.085	-.166	-.139	-1.412	-.121	-.710	-.048	-.663
size.impact.cor.schoolL: sd.schoolL	1.442	1.268	.029	1.808.	1.551.	.336	.281	1.306	.005	.271	-.332	.303
size.impact.cor.schoolL: ICC.schoolL	1.866.	1.821.	.059	1.332	1.446.	.197	.167	1.727.	.024	.687	.537*	.641
size.impact.cor.schoolL: variable.nTRUE	.302	.238	.196*	.750	.658	.658**	.588**	.269	.136	.295	-.185	.287
size.impact.cor.schoolL: cv.schoolS	2.144.	2.058.	-.160.	2.125.	2.088.	.035	-.008	2.041.	-.177.	1.714.	.243	1.707.
size.impact.cor.schoolL: cv.schoolL	1.479	1.430	-.066	2.170.	2.032.	.430.	.369	1.495	-.140	1.760.	.505	1.766.
size.impact.cor.schoolL: num.cov.schoolS	-1.287	-1.223	.005	-1.046	-.909	-.058	-.014	-1.233	.007	-.836	-.330	-.805
size.impact.cor.schoolL: num.cov.schoolL	-.632	-.660	-.091	-.243	-.218	-.113	-.059	-.699	-.110	.551	-.221	.571
sd.schoolL: ICC.schoolL	9.881***	10.098***	.112	9.106***	9.813***	-.117	-.063	9.915***	-.059	8.586***	-.285	8.563***
sd.schoolL: variable.nTRUE	.529	.677	-.171*	1.716.	1.278	.093	-.046	.910	.091	.513	-.037	.551
sd.schoolL: cv.schoolS	-2.729*	-2.534*	-.085	-1.845	-1.948.	-.094	-.167	-2.455*	.008	-1.367	-.384	-1.271
sd.schoolL: cv.schoolL	-.079	.002	-.105	.818	.318	.301	.164	.274	.196.	.439	-.139	.497
sd.schoolL: num.cov.schoolS	1.332	1.318	.017	1.329	1.496	.007	-.047	1.298	<.001	.506	.175	.551
sd.schoolL: num.cov.schoolL	-.515	-.535	.017	-.220	-.177	-.064	-.058	-.514	.008	-.597	-.227	-.564
ICC.schoolL: variable.nTRUE	4.219***	4.340***	-.196*	5.149***	4.426***	.073	-.092	4.945***	.736***	2.511***	.120	2.483***
ICC.schoolL: cv.schoolS	.260	.423	-.126	1.464	1.186	-.133	-.289	.582	.085	-.177	.366	-.175
ICC.schoolL: cv.schoolL	3.902**	4.208***	-.396***	4.093***	3.573***	.014	-.076	4.956***	.791***	2.307*	.248	2.294*
ICC.schoolL: num.cov.schoolS	.304	.340	-.028	.614	.599	-.196	-.201	.380	-.008	-.124	-.072	-.111
ICC.schoolL: num.cov.schoolL	-.217	-.213	-.015	.056	.119	-.095	-.089	-.207	-.041	.447	.481	.477
variable.nTRUE: cv.schoolS	-1.044	-1.117	.025	-2.644*	-2.246*	3.750***	2.253***	-1.277	.360***	-1.640.	-.550.	-1.635.
variable.nTRUE: cv.schoolL	-5.616***	-5.661***	.568***	-6.344***	-5.707***	10.657***	6.163***	-5.877***	2.479***	-4.975***	-.499	-4.965***
variable.nTRUE: num.cov.schoolS	.188	.204	-.098	.141	.099	-.036	-.027	.209	-.070	.551	.231	.521
variable.nTRUE: num.cov.schoolL	-.281	-.241	.109	-.594	-.606	.256	.266	-.331	.079	-.661	.268	-.749
cv.schoolS: num.cov.schoolS	.342	.437	.155	.440	.470	.135	.145	.442	.158	-.602	.024	-.606
cv.schoolL: num.cov.schoolS	1.890	1.844	.041	1.722	1.653	.154	.158	1.846	.081	1.122	.073	1.086
cv.schoolS: num.cov.schoolL	1.944	1.886	.112	1.703	1.594	.122	.123	1.872	.105	.315	-.158	.311
cv.schoolL: num.cov.schoolL	2.079	1.980	-.048	1.849	1.724	.162	.176	1.858	-.086	.719	.262	.709

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Most SE Estimate Inaccuracy by Method

method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	2.53	S	0	0	S	L	0	L
LRa-FE-db	2.39	0	0	0	0	0	0	L
LRa-FE-db	2.28	0	0	0	L	L	0	L
LRa-FE-db	2.19	S	0	0	L	L	0	L
LRa-FE-db	2.13	0	0	0	S	L	L	L
LRa-FE-het	2.55	S	0	0	S	L	0	L
LRa-FE-het	2.39	0	0	0	0	0	0	L
LRa-FE-het	2.32	0	0	0	L	L	0	L
LRa-FE-het	2.21	S	0	0	L	L	0	L
LRa-FE-het	2.17	0	0	0	S	L	0	L
LRapw-FE-db	2.39	0	0	0	0	0	0	L
LRapw-FE-db	2.28	0	0	0	L	L	0	L
LRapw-FE-db	2.19	S	0	0	L	L	0	L
LRapw-FE-db	2.14	S	0	0	S	L	0	L
LRapw-FE-db	2.13	0	0	0	S	L	0	L
LRapw-FE-het	2.39	0	0	0	0	0	0	L
LRapw-FE-het	2.32	0	0	0	L	L	0	L
LRapw-FE-het	2.21	S	0	0	L	L	0	L
LRapw-FE-het	2.19	S	0	0	S	L	0	L
LRapw-FE-het	2.17	0	0	0	S	L	0	L
LRicw-FE-crve	2.54	S	0	0	S	L	0	L
LRicw-FE-crve	2.39	0	0	0	0	0	0	L
LRicw-FE-crve	2.32	0	0	0	L	L	0	L
LRicw-FE-crve	2.21	S	0	0	L	L	0	L

method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRicw-FE-crve	2.17	0	0	0	S	L	0	L
MLM-FE	2.02	L	0	0	S	0	0	L
MLM-FE	1.97	S	0	0	L	S	L	L
MLM-FE	1.9	S	0	0	0	L	0	L
MLM-FE	1.89	L	0	0	0	S	0	L
MLM-FE	1.89	L	0	0	S	L	L	L
MLM-FIcw	1.27	L	L	0	0	S	0	L
MLM-FIcw	1.25	0	L	0	0	0	0	L
MLM-FIcw	1.25	L	L	0	0	L	0	L
MLM-FIcw	1.25	L	L	0	S	S	L	L
MLM-FIcw	1.25	L	L	0	S	0	L	L
MLM-RE	2.03	L	0	0	S	0	0	L
MLM-RE	1.97	S	0	0	L	S	L	L
MLM-RE	1.9	S	0	0	0	L	0	L
MLM-RE	1.89	L	0	0	0	S	0	L
MLM-RE	1.89	L	0	0	S	L	L	L
LRapw-FIpw-db	0.671	0	0	0	L	L	0	L
LRapw-FIpw-db	0.714	0	0	0	L	0	L	L
LRapw-FIpw-db	0.714	0	0	0	L	S	0	L
LRapw-FIpw-db	0.716	S	0	0	L	L	L	L
LRapw-FIpw-db	0.723	L	0	0	L	S	0	L
LRapw-FIpw-het	0.748	0	0	0	L	L	0	L
LRapw-FIpw-het	0.756	S	0	0	L	L	L	L
LRapw-FIpw-het	0.761	L	0	0	L	S	0	L
LRapw-FIpw-het	0.767	0	0	0	L	S	0	L
LRapw-FIpw-het	0.781	0	0	0	L	S	L	L

The Effect of Different Data Generating Parameters on the Estimate Standard Error Stability

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	.009***	.009***	.010***	.010***	.010***	.010***	.010***	.009***	.010***	.006***	.006***	.006***
block_typeS	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	<.001	.001
block_typeL	.006***	.006***	.003**	.005***	.005***	.004***	.004**	.006***	.003**	.006***	.001	.006***
J.barS	-.003**	-.003**	-.003**	-.003*	-.002.	-.003**	-.003**	-.003**	-.003**	-.002*	-.002*	-.002*
J.barL	-.005***	-.005***	-.006***	-.004***	-.003**	-.006***	-.006***	-.005***	-.006***	-.004***	-.004***	-.004***

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
size.impact.cor.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001
sd.schoolL	.004***	.004***	.005***	.005***	.004***	.005***	.005***	.004***	.005***	.007***	.009***	.007***
ICC.schoolL	.012***	.012***	.014***	.011***	.011***	.013***	.013***	.012***	.014***	.015***	.017***	.015***
variable.nTRU E	.002*	.002*	.002.	-.001	-.001	<.001	<.001	.002*	.002*	.001	.001	.001
cv.schoolS	.001	<.001	<.001	<.001	<.001	<.001	<.001	.001	<.001	<.001	<.001	<.001
cv.schoolL	.002*	.002.	.002.	-.002.	-.002.	-.001	-.001	.002*	.002*	<.001	<.001	<.001
num.cov.schoo lS	-.001	-.001	-.001	<.001	<.001	-.001	-.001	-.001	-.001	<.001	-.001	<.001
num.cov.schoo lL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeS: J.barS	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	<.001	<.001	<.001
block_typeL: J.barS	<.001	<.001	.001	<.001	-.001	<.001	.001	<.001	.001	-.001	<.001	-.001
block_typeS: J.barL	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	<.001	<.001	<.001
block_typeL: J.barL	-.003***	-.003***	-.002*	-.004***	-.004***	-.002*	-.001.	-.003***	-.002.	-.003***	-.001	-.003***
block_typeS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: size.impact.cor.schoolL	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001
block_typeS: sd.schoolL	<.001	.001	.001	<.001	<.001	.001	.001	.001	.001	<.001	<.001	<.001
block_typeL: sd.schoolL	.001	.001	.001	<.001	.001	.001	.001	.001	.001	-.001.	.001	-.001.
block_typeS: ICC.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: ICC.schoolL	.004***	.004***	.005***	.003***	.004***	.005***	.005***	.004***	.005***	.001	.003***	.001
block_typeS: variable.nTRUE	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: variable.nTRUE	<.001	<.001	.001	.003***	.004***	<.001	<.001	<.001	.001	<.001	.001	<.001
block_typeS: cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: cv.schoolS	.001	.001	.001	.002.	.002.	.001	.001	.001	.001	<.001	<.001	<.001
block_typeS: cv.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: cv.schoolL	.001	.001	.002**	.004***	.005***	.001	.001	.001	.002*	.001	.001	.001
block_typeS: num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: num.cov.schoolS	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeS: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: num.cov.schoolL	-.001	-.001	-.001	<.001	<.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001
J.barS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	-.001	<.001
J.barL: size.impact.cor.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001.	-.001*	-.001.
J.barS: sd.schoolL	<.001	<.001	-.001	-.001	-.001	-.001	-.001	<.001	-.001	-.001*	-.002**	-.001*
J.barL: sd.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.002***	-.003***	-.002***
J.barS: ICC.schoolL	-.004***	-.004***	-.004***	-.004***	-.005***	-.005***	-.005***	-.004***	-.004***	-.005***	-.005***	-.005***
J.barL: ICC.schoolL	-.006***	-.006***	-.007***	-.008***	-.008***	-.008***	-.008***	-.006***	-.007***	-.007***	-.009***	-.007***
J.barS: variable.nTRUE	<.001	<.001	<.001	<.001	-.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL: variable.nTRUE	-.001*	-.001*	-.002*	-.002*	-.003***	-.001	-.001.	-.001*	-.002*	-.001	-.001.	-.001
J.barS: cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL: cv.schoolS	<.001	<.001	<.001	-.001	-.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barS: cv.schoolL	-.001.	-.001.	-.002.	-.001	-.002*	-.001	-.001	-.001.	-.002.	-.001	-.001	-.001
J.barL: cv.schoolL	-.002*	-.002*	-.002**	-.002*	-.003***	-.001	-.002.	-.002*	-.002**	<.001	<.001	<.001
J.barS: num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	.001	<.001
J.barL: num.cov.schoolS	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001
J.barS: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	-.001	-.001	<.001	<.001	<.001	<.001	<.001
J.barL: num.cov.schoolL	<.001	<.001	-.001	-.001	-.001	-.001	-.001	<.001	-.001	<.001	<.001	<.001
size.impact.cor.schoolL: sd.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
size.impact.cor.schoolL: ICC.schoolL	.001*	.001*	.001.	.001*	.001*	.001.	.001.	.001*	.001.	.001.	.001.	.001.
size.impact.cor.schoolL: variable.nTRUE	.001*	.001*	.001*	.001	.001.	.001.	.001.	.001*	.001*	.001*	.001*	.001*
size.impact.cor.schoolL: cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
size.impact.cor.schoolL: cv.schoolL	.001	.001	<.001	<.001	.001	<.001	<.001	.001	<.001	<.001	<.001	<.001
size.impact.cor.schoolL: num.cov.schoolS	<.001	<.001	.001	<.001	<.001	.001	.001	<.001	.001	<.001	<.001	<.001
size.impact.cor.schoolL: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	.001	.001	<.001	.001	.001	.001	.001
sd.schoolL: ICC.schoolL	-.001**	-.001**	-.002*	-.002***	-.002**	-.002***	-.002***	-.001**	-.001*	-.003***	-.005***	-.003***

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
sd.schoolL: variable.nTRUE	-.001*	-.001*	-.001*	-.001	<.001	<.001	<.001	-.001*	-.001*	-.001.	-.001	-.001
sd.schoolL: cv.schoolS	-.002**	-.002**	-.002*	-.002*	-.002*	-.002*	-.001*	-.002**	-.002*	-.001*	-.001.	-.001.
sd.schoolL: cv.schoolL	-.001*	-.001*	-.001*	<.001	<.001	<.001	<.001	-.001*	-.001*	-.001	<.001	<.001
sd.schoolL: num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
sd.schoolL: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
ICC.schoolL: variable.nTRUE	-.001*	-.001*	-.001*	.003***	.003***	.002***	.003***	-.002**	-.002**	<.001	<.001	<.001
ICC.schoolL: cv.schoolS	<.001	<.001	<.001	.001	.001	.001	.001	<.001	<.001	<.001	<.001	<.001
ICC.schoolL: cv.schoolL	-.003***	-.003***	-.003***	.003***	.003***	.002*	.002**	-.003***	-.003***	<.001	-.001	-.001
ICC.schoolL: num.cov.schoolS	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001
ICC.schoolL: num.cov.schoolL	<.001	<.001	-.001	<.001	<.001	-.001	-.001	<.001	-.001	<.001	<.001	<.001
variable.nTRUE: cv.schoolS	<.001	<.001	<.001	.002*	.002**	<.001	.001	<.001	<.001	<.001	<.001	<.001
variable.nTRUE: cv.schoolL	.004***	.004***	.005***	.006***	.008***	.003***	.005***	.005***	.006***	.001	.001*	.001
variable.nTRUE: num.cov.schoolS	-.001*	-.001*	-.002*	-.001*	-.002*	-.002*	-.002*	-.001*	-.002*	-.001*	-.001.	-.001*
variable.nTRUE: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
cv.schoolS: num.cov.schoolS	.001.	.001.	.002*	.001	.001	.002*	.002*	.001.	.002*	.001	.001	.001
cv.schoolL: num.cov.schoolS	.002*	.002*	.002*	.002*	.002*	.002*	.002*	.002*	.002*	.001*	.001*	.001*
cv.schoolS: num.cov.schoolL	<.001	<.001	.001	<.001	<.001	.001	.001	<.001	.001	<.001	<.001	<.001
cv.schoolL: num.cov.schoolL	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Most SE Estimate Instability by Method

method	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRapw-FE-db	0.0585	0	L	L	L	0	L	L
LRapw-FE-db	0.0585	0	L	L	L	L	L	L
LRapw-FE-db	0.0582	0	0	L	L	L	L	L
LRapw-FE-db	0.0573	0	0	L	L	0	L	L
LRapw-FE-db	0.0569	0	0	L	S	L	L	L
LRapw-FE-het	0.0717	0	L	L	L	0	L	L
LRapw-FE-het	0.0715	0	L	L	L	L	L	L
LRapw-FE-het	0.0715	0	0	L	L	L	L	L
LRapw-FE-het	0.0673	0	0	L	L	0	L	L
LRapw-FE-het	0.0638	0	L	L	L	0	0	L
MLM-FIcw	0.0519	0	L	L	0	L	L	L
MLM-FIcw	0.0516	0	L	L	0	0	L	L

method	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FIcw-db	0.0659	S	0	L	S	S	0	L
LRa-FIcw-db	0.0612	0	L	L	L	L	L	L
LRa-FIcw-db	0.056	0	L	L	S	S	0	L
LRa-FIcw-db	0.0541	0	L	L	L	0	L	L
LRa-FIcw-db	0.0519	0	L	L	S	0	0	L
LRapw-FIpw-db	0.0659	S	0	L	S	S	0	L
LRapw-FIpw-db	0.0565	0	L	L	L	L	L	L
LRapw-FIpw-db	0.056	0	L	L	S	S	0	L
LRapw-FIpw-db	0.0519	0	L	L	S	0	0	L
LRapw-FIpw-db	0.0519	0	L	L	0	L	L	L
LRapw-FIpw-het	0.0659	S	0	L	S	S	0	L
LRapw-FIpw-het	0.0579	0	L	L	L	L	L	L
LRapw-FIpw-het	0.056	0	L	L	S	S	0	L
LRapw-FIpw-het	0.0521	0	L	L	L	0	L	L
LRapw-FIpw-het	0.0519	0	L	L	S	0	0	L
LRicw-FIcw-crve	0.0659	S	0	L	S	S	0	L
LRicw-FIcw-crve	0.0613	0	L	L	L	L	L	L
LRicw-FIcw-crve	0.056	0	L	L	S	S	0	L
LRicw-FIcw-crve	0.0543	0	L	L	L	0	L	L
LRicw-FIcw-crve	0.0519	0	L	L	S	0	0	L

Top Five Scenarios with Most Normalized SE Estimate Instability by Method

method	sd.SE hat.s.norm	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	2.1	0.01	L	0	0	L	S	0	0
LRa-FE-db	2.06	0.00985	L	0	0	L	L	0	0

method	sd.SE hat.s.norm	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	2.04	0.00986	L	0	0	L	L	L	0
LRa-FE-db	2.03	0.00951	L	0	0	L	L	L	S
LRa-FE-db	2.03	0.00791	L	0	0	L	L	0	S
LRa-FE-het	2.1	0.01	L	0	0	L	S	0	0
LRa-FE-het	2.06	0.00985	L	0	0	L	L	0	0
LRa-FE-het	2.04	0.00986	L	0	0	L	L	L	0
LRa-FE-het	2.03	0.00792	L	0	0	L	L	0	S
LRa-FE-het	2.03	0.00951	L	0	0	L	L	L	S
LRapw-FE-db	1.49	0.0198	0	L	0	S	S	0	L
LRapw-FE-db	1.47	0.0155	S	0	0	S	L	0	L
LRapw-FE-db	1.44	0.0122	L	0	L	L	L	0	L
LRapw-FE-db	1.42	0.0247	S	0	0	S	0	L	L
LRapw-FE-db	1.41	0.0197	L	0	L	L	0	L	L
LRapw-FE-het	1.52	0.0337	0	0	L	L	S	0	L
LRapw-FE-het	1.51	0.0361	L	0	L	L	S	0	L
LRapw-FE-het	1.49	0.0127	L	0	L	L	L	0	L
LRapw-FE-het	1.48	0.0589	0	0	L	L	L	0	L
LRapw-FE-het	1.48	0.0206	L	0	L	L	0	L	L
LRicw-FE-crve	2.16	0.0103	L	0	0	L	S	0	0
LRicw-FE-crve	2.12	0.0102	L	0	0	L	L	0	0
LRicw-FE-crve	2.1	0.0102	L	0	0	L	L	L	0
LRicw-FE-crve	2.09	0.0098	L	0	0	L	L	L	S
LRicw-FE-crve	2.08	0.00812	L	0	0	L	L	0	S
MLM-FIcw	1.15	0.0148	0	L	0	0	0	0	0
MLM-FIcw	1.15	0.0146	0	L	0	0	0	L	0
MLM-FIcw	1.15	0.0148	0	L	0	S	0	L	0

method	sd.SE hat.s.norm	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
MLM-FIcw	1.15	0.0148	0	L	0	0	S	0	0
MLM-FIcw	1.15	0.0147	0	L	0	0	L	L	0
LRa-FIcw-db	2.12	0.0101	L	0	0	L	S	0	0
LRa-FIcw-db	2.07	0.00991	L	0	0	L	L	0	0
LRa-FIcw-db	2.06	0.00995	L	0	0	L	L	L	0
LRa-FIcw-db	2.05	0.00801	L	0	0	L	L	0	S
LRa-FIcw-db	2.05	0.00962	L	0	0	L	L	L	S
LRapw-FIpw-db	1.38	0.0388	S	L	0	0	0	L	L
LRapw-FIpw-db	1.34	0.0659	S	0	L	S	S	0	L
LRapw-FIpw-db	1.33	0.0369	0	0	L	0	S	L	L
LRapw-FIpw-db	1.33	0.0143	L	L	L	L	L	0	0
LRapw-FIpw-db	1.33	0.0268	L	0	L	L	S	L	L
LRapw-FIpw-het	1.42	0.0152	L	L	L	L	L	0	0
LRapw-FIpw-het	1.41	0.0154	L	L	L	L	0	0	0
LRapw-FIpw-het	1.39	0.0152	L	L	L	L	S	0	0
LRapw-FIpw-het	1.39	0.0177	L	L	L	L	L	0	S
LRapw-FIpw-het	1.38	0.014	L	0	L	L	0	L	0
LRicw-FIcw-crve	2.21	0.0106	L	0	0	L	S	0	0
LRicw-FIcw-crve	2.17	0.0104	L	0	0	L	L	0	0
LRicw-FIcw-crve	2.15	0.0104	L	0	0	L	L	L	0
LRicw-FIcw-crve	2.14	0.01	L	0	0	L	L	L	S
LRicw-FIcw-crve	2.13	0.00831	L	0	0	L	L	0	S

Are the estimated standard errors targeting the finite or super population

*This section answers the questions on page 8.

*As long as the school level ICC isn't zero, than the true finite and super population standard errors are very similar to each other (graph 2).

*I was having a little trouble visualizing whether the standard errors were targeting the finite population or super population standard errors, so I made up this metric: difference in % inaccuracy. This is calculated as the magnitude of ((Mean Estimated SE - True Super Population SE) / True Super Population SE) x 100 minus the magnitude of ((Mean Estimated SE - True Finite Population SE) / True Finite Population SE) x 100. When this number is positive the estimated standard errors are closer to the true super population standard errors and when this number is negative the estimated standard errors are closer to the finite population.

*The only estimators with standard errors that are ever really significantly closer to the finite standard errors than the super population standard errors are the person weighted fully interacted one's and the fully interacted MLMs (graphs three and four). It does look like the person weighted fully interacted estimators are targeting the finite standard errors when the ICC is zero, the blocks are very dissimilar, and there is high variance in the cluster sizes. Otherwise these estimators are also targetting super population standard errors (graph five).

Description of Analysis:

*The first graph is a boxplot of ratio of the true finite standard errors over the true super population standard errors across the different scenarios by method. When this metric is one the two standard errors are the same, when it's above one the finite population standard errors are bigger and when it's below one the super population standard errors are bigger.

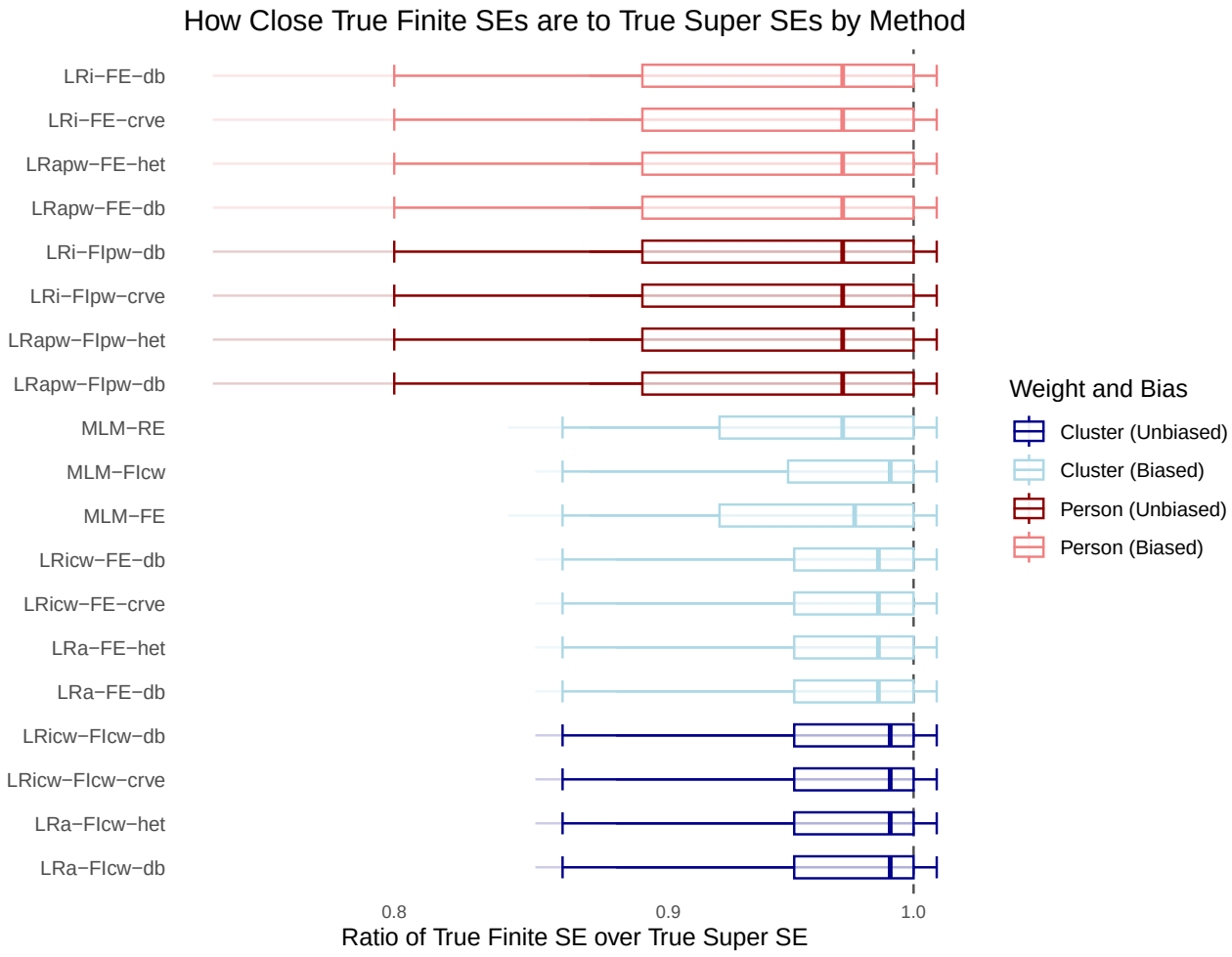
*The second graph is the same as the first but broken out by ICC. In the graph on the left the school level ICC is zero and on the right it's .20.

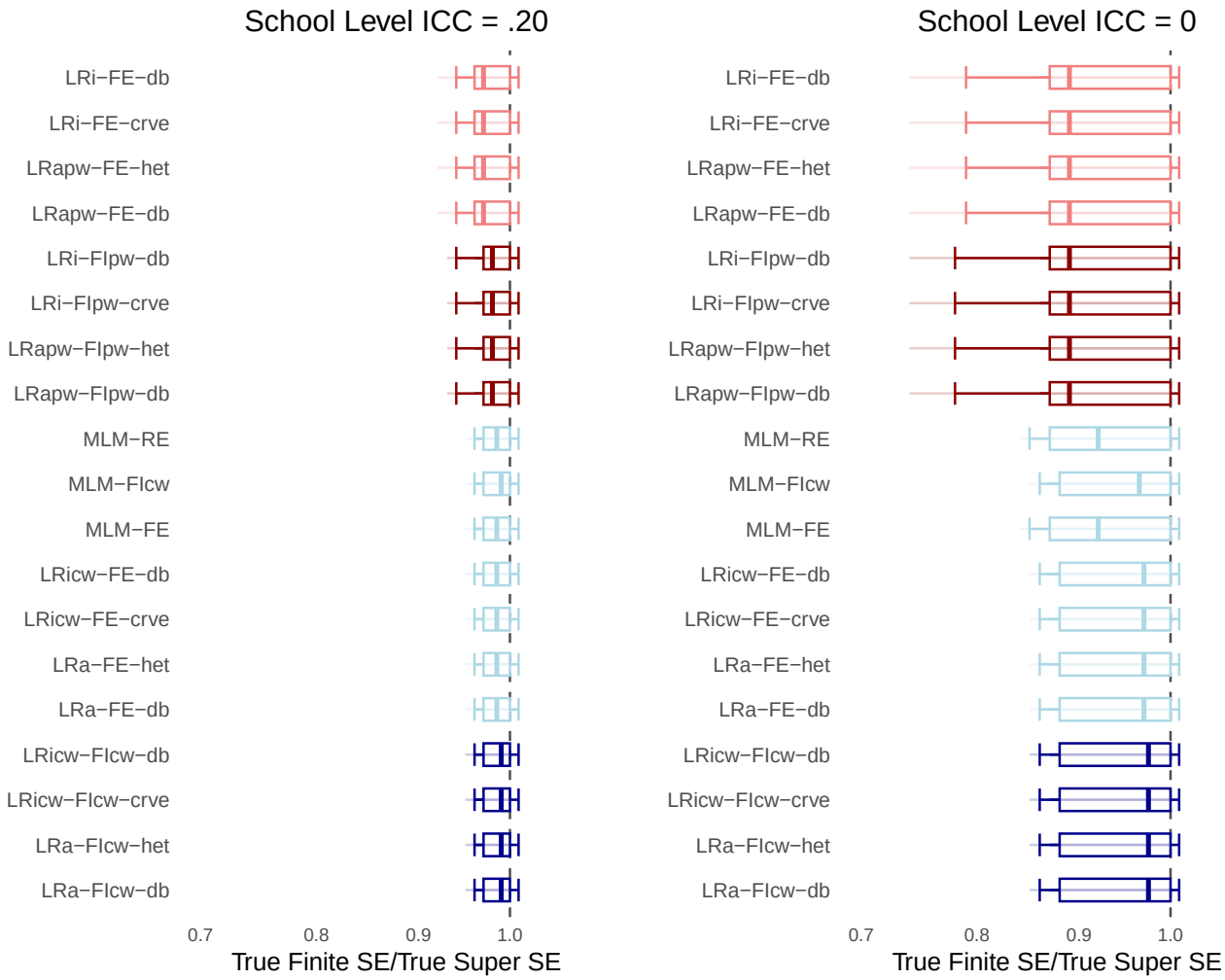
*The third graph is a boxplot of the difference in % inaccuracy across methods. This is calculated as the magnitude of ((Mean Estimated SE - True Super Population SE) / True Super Population SE) x 100 minus the magnitude of ((Mean Estimated SE - True Finite Population SE) / True Finite Population SE) x 100. When this number is positive the estimated standard errors are closer to the true super population standard errors and when this number is negative the estimated standard errors are closer to the finite population.

*The fourth graph is the same as the third but broken out by ICC. In the graph on the left the school level ICC is zero and on the right it's .20.

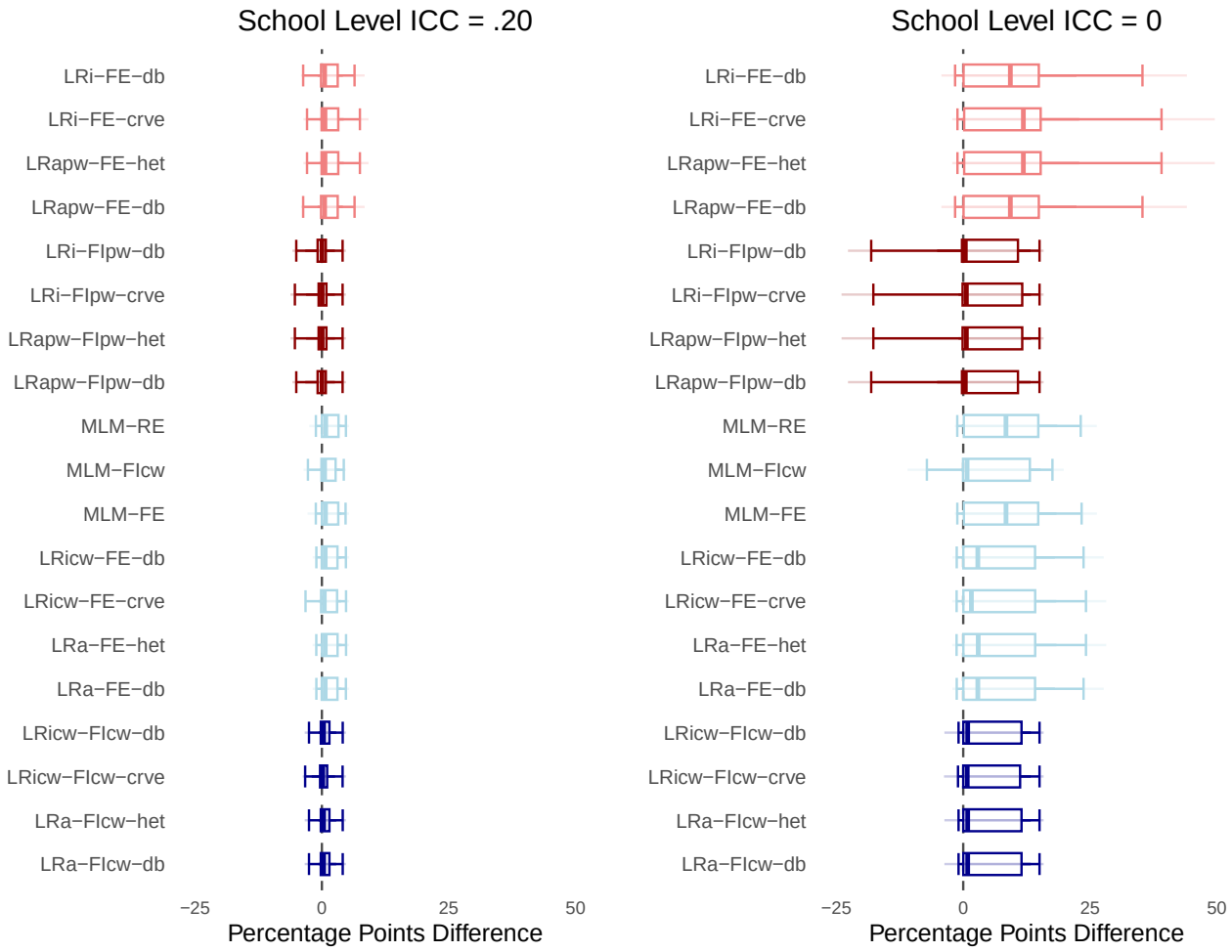
*The fifth graph is a boxplot of the difference in % inaccuracy across methods, but only for the methods that might be targeting the finite population standard errors and only when the school level ICC is zero, the blocks are all messy, and the cluster size coefficient of variation is large.

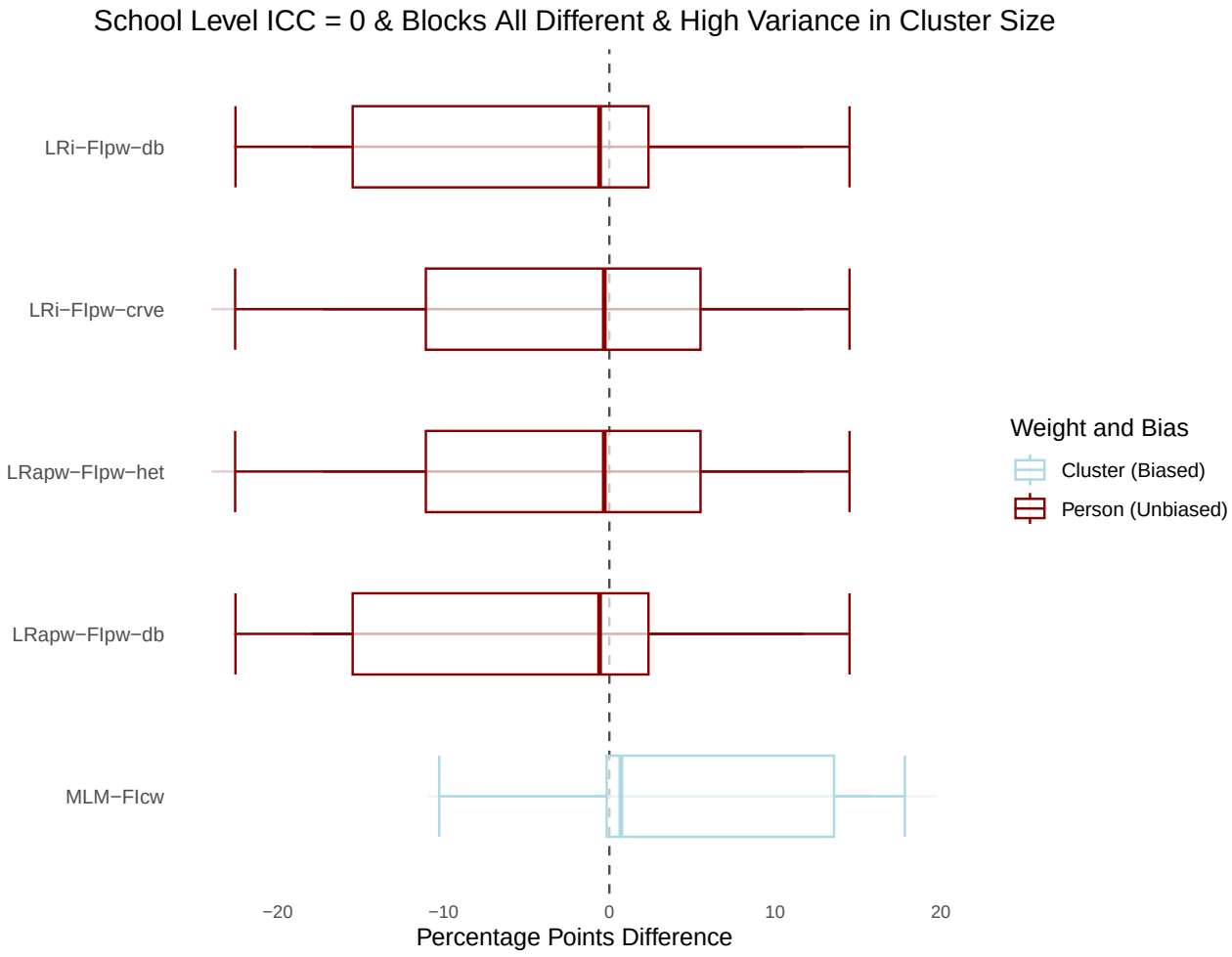
*The table contains the five DGPs that produce the largest difference in % inaccuracy for each method, conditional on the magnitude being at least five percentage points.


How Close True Finite SEs are to True Super SEs by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.8
0.9
1.0
Ratio of True Finite SE over True Super SE
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)


School Level ICC = .20
School Level ICC = 0
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.7
0.8
0.9
1.0
True Finite SE/True Super SE


How Much Closer the Estimated S.E. is to the True Super Population SE by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
−25
0
25
50
Percentage Points Difference
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)


School Level ICC = .20
School Level ICC = 0
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
−25
0
25
50
Percentage Points Difference



Top Five Scenarios with Where Estimated SEs are Closest to Targeting Finite SEs

method	dif	infla te.f	infla te.s	true se.ratio	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
MLM-FIcw	-11	0.986	0.877	0.89	L	L	0	L	L	0	L
MLM-FIcw	-9.98	0.978	0.878	0.9	S	L	0	L	L	L	L
MLM-FIcw	-9.74	0.963	0.866	0.9	S	L	0	L	0	0	L
MLM-FIcw	-9.57	0.973	0.877	0.9	0	L	0	S	S	0	L
MLM-FIcw	-8.31	0.928	0.844	0.91	S	L	0	0	0	L	L
LRapw-FIpw-db	-22.7	0.943	0.716	0.76	S	0	0	L	L	L	L
LRapw-FIpw-db	-22.5	1.01	0.766	0.76	0	L	0	L	L	L	L
LRapw-FIpw-db	-22.2	0.987	0.765	0.78	S	L	0	L	S	0	L
LRapw-FIpw-db	-21.4	1.02	0.763	0.75	0	L	0	L	0	0	L
LRapw-FIpw-db	-20.9	0.933	0.723	0.78	L	0	0	L	S	0	L
LRapw-FIpw-het	-24	0.996	0.756	0.76	S	0	0	L	L	L	L
LRapw-FIpw-het	-22	0.981	0.761	0.78	L	0	0	L	S	0	L

method	dif	infla te.f	infla te.s	true se.ratio	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRapw-FIpw-het	-21	0.977	0.767	0.79	0	0	0	L	S	0	L
LRapw-FIpw-het	-19.1	0.976	0.785	0.8	0	0	0	L	0	0	L
LRapw-FIpw-het	-18.7	0.968	0.781	0.81	0	0	0	L	S	L	L

What do the confidence intervals look like for each estimator

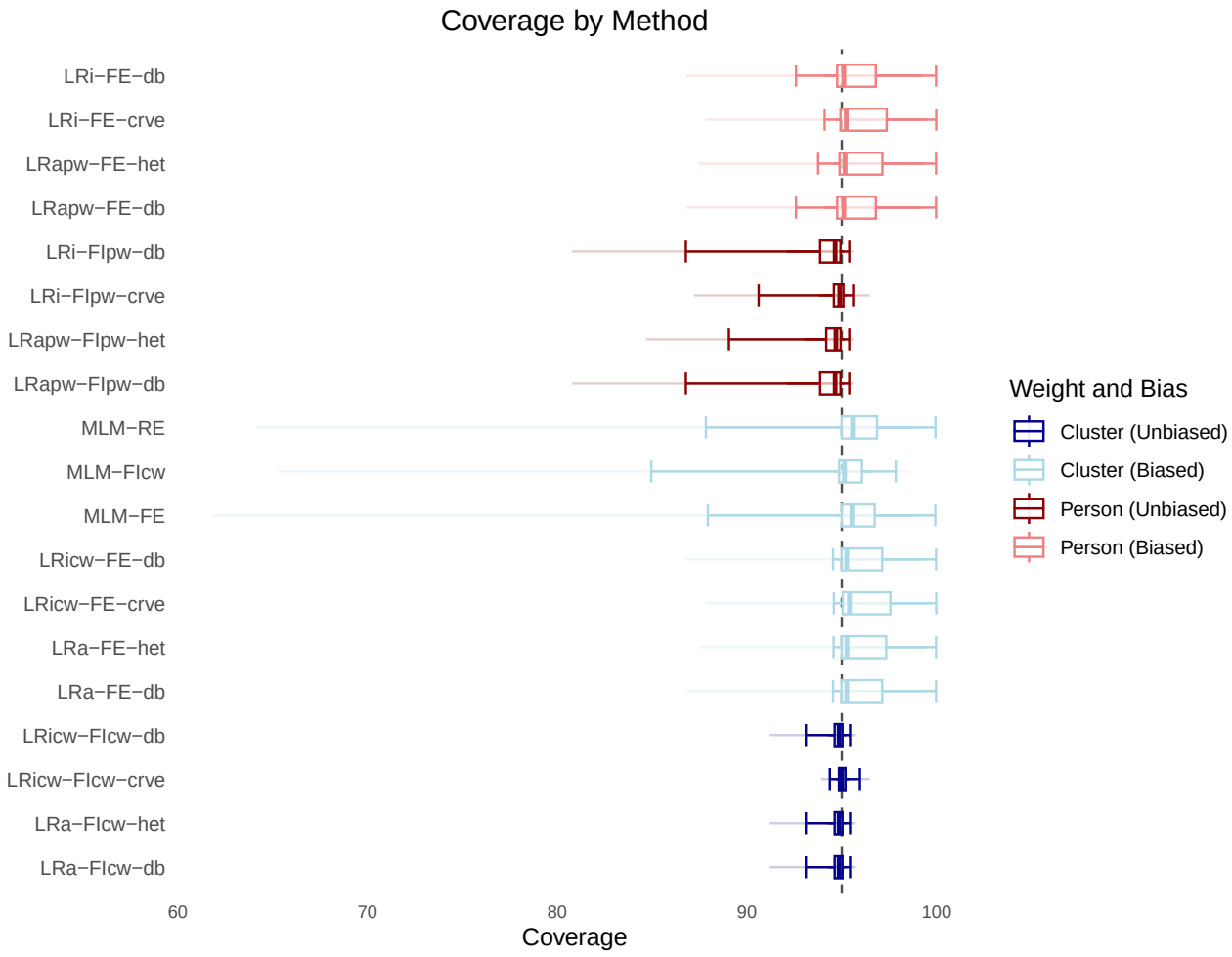
Main Bullet Points:

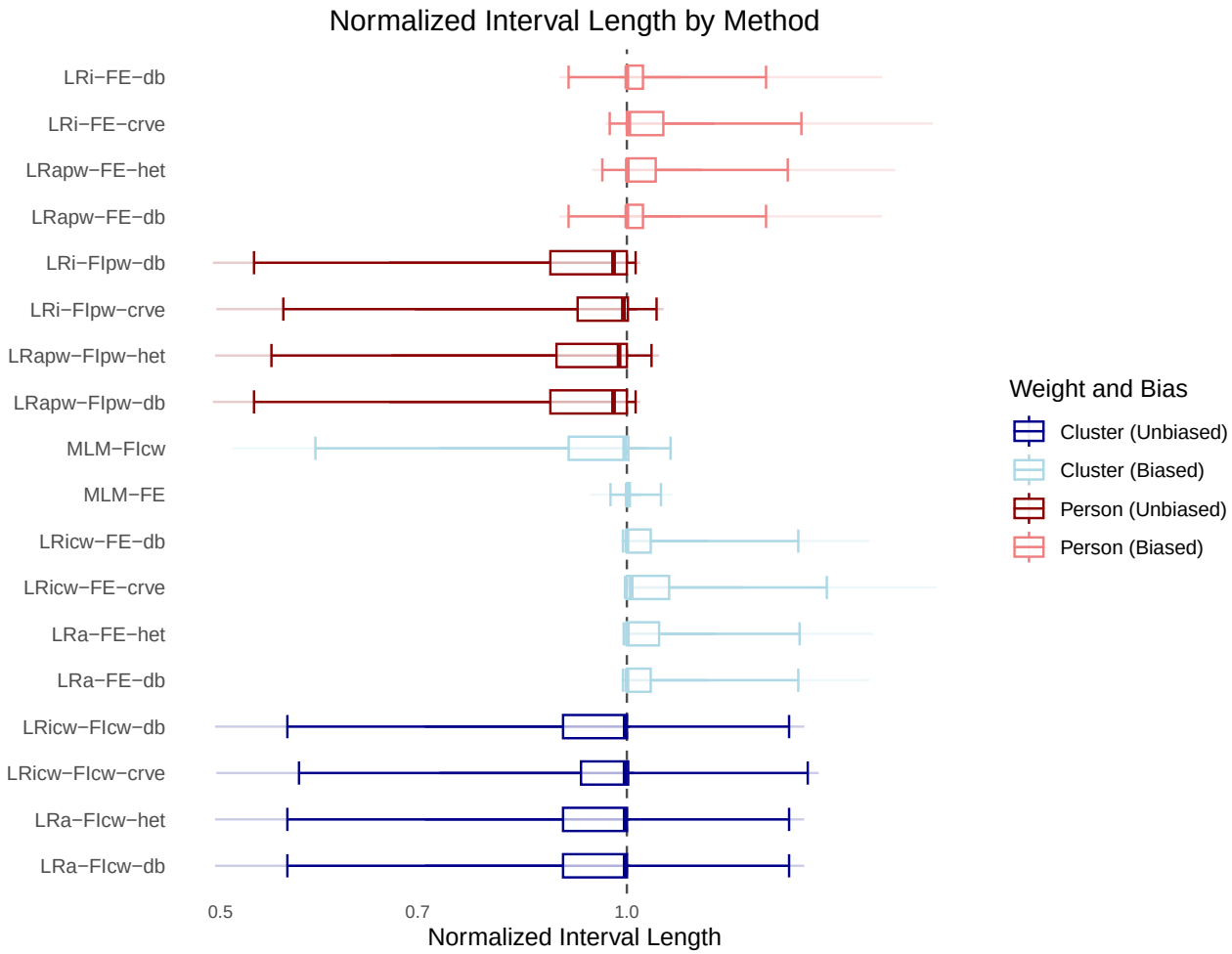
*Coverage is the percentage of simulation replications where the estimator is within the 95% confidence interval. We would like the number to 95% (the boxplots have a line there). The notable things are that the coverage for the MLM estimators is often very bad but other than that the coverage for the cluster weighted estimators is generally good. The coverage rates for the person weighted estimators however are pretty bad across methods,it's either very low (meaning the SEs are being underestimated) or too high (meaning the SEs are being overestimated within the same methods.

Description of Analysis:

*The first graph is a boxplot of the coverage by method. Coverage is the percentage of treatment effect estimates that fall in the 95% confidence interval. The vertical line is at 95%, which is what coverage should be.

*The second graph is a boxplot of the normalized 95% confidence interval length by method.


Coverage by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
60
70
80
90
100
Coverage
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)



Is Low Degrees of Freedom Causing Problems because of Estimate Instability Driven or Bias

Main Bullet Points:

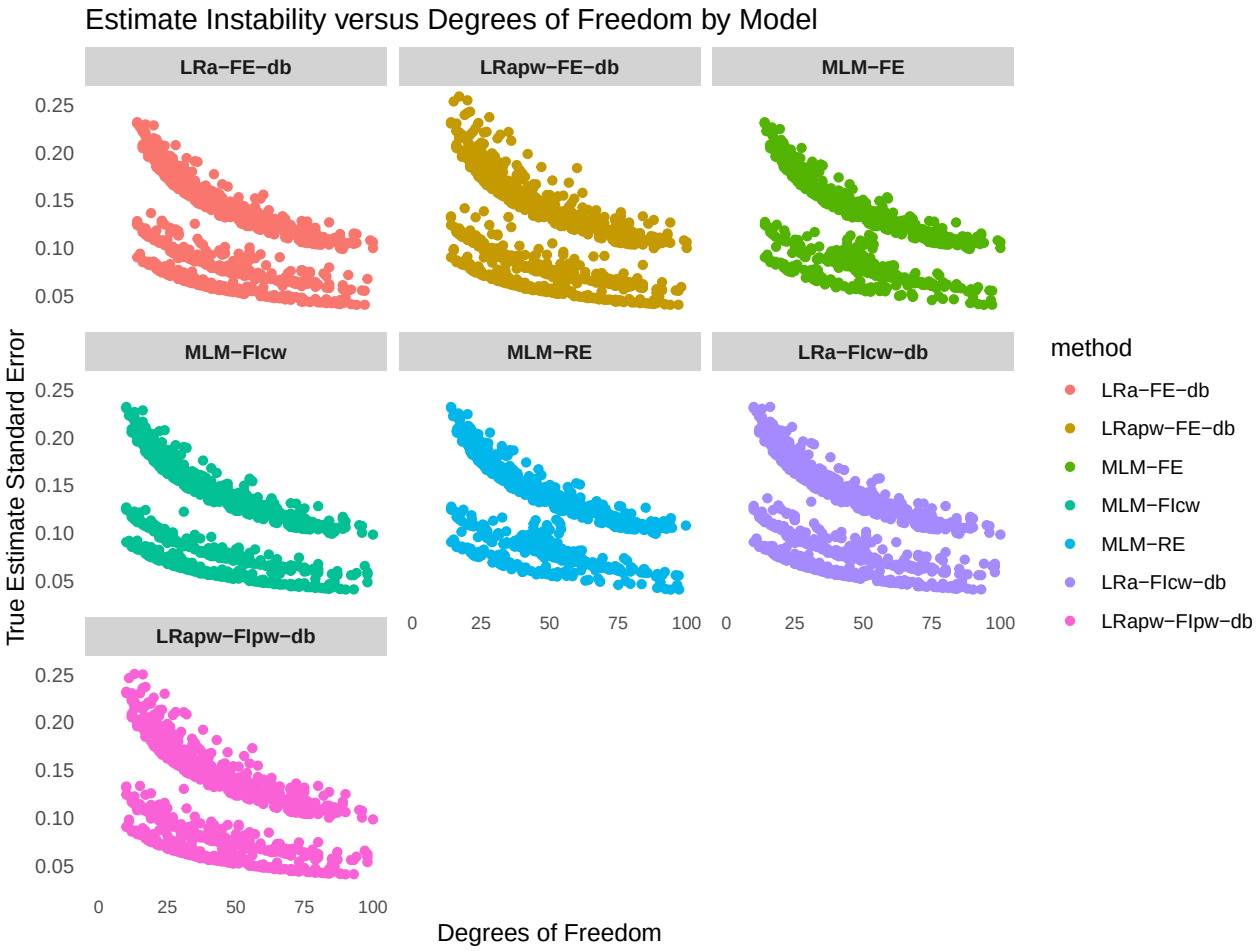
*This is answering the question on page 35

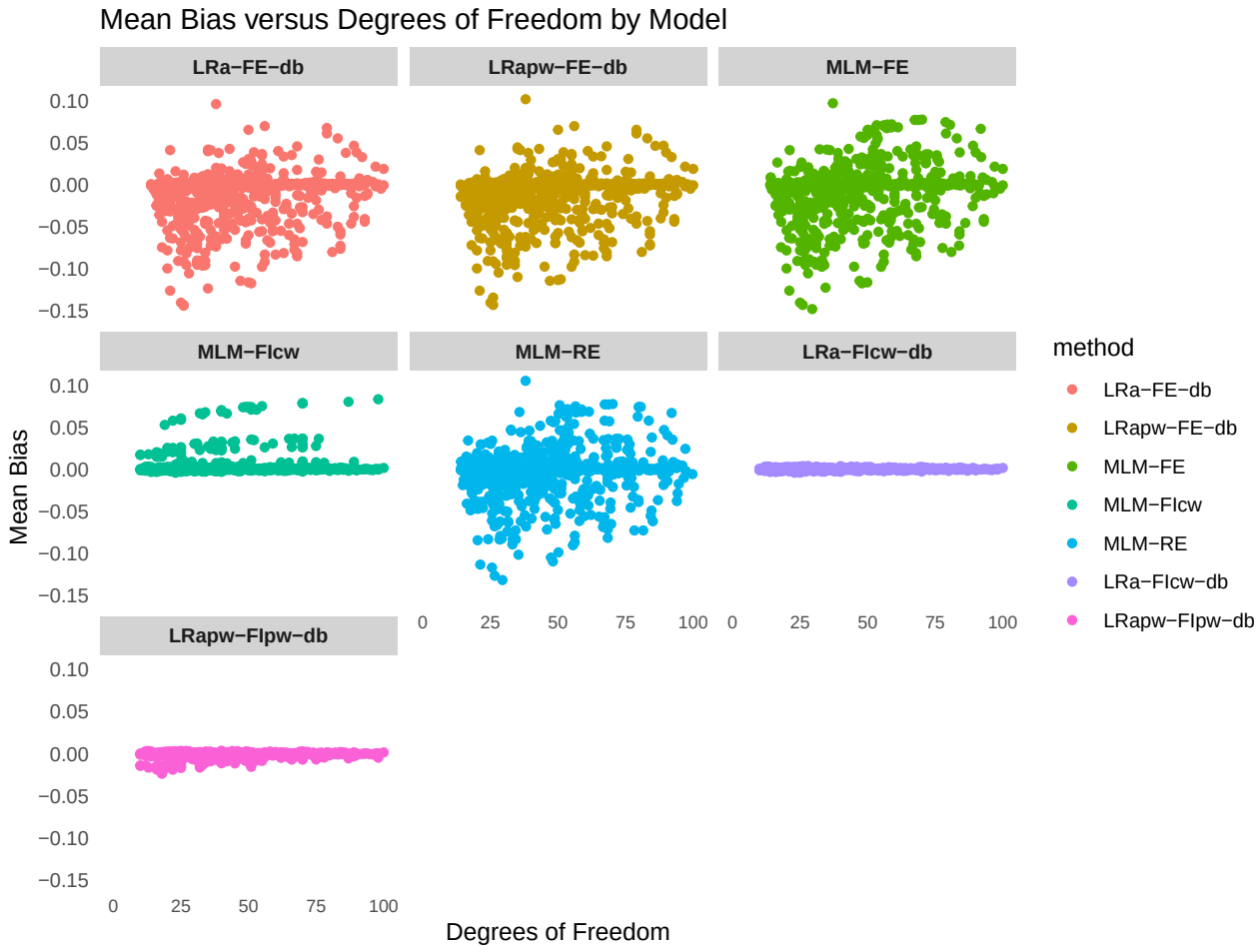
*Low degrees of freedom is clearly causing problems because of estimate instability. There is clear increase in estimate instability for the low degrees of freedom DGPs (graph one) and no relation between degrees of freedom and mean bias (graph two).

Description of Analysis:

*The first graph is scatter plot of the true estimate standard error across replications against the model degrees of freedom, graphed sepparately for each method

*The second graph is scatter plot of the mean bias across replications against the model degrees of freedom, graphed sepparately for each method


Estimate Instability versus Degrees of Freedom by Model
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db
True Estimate Standard Error
0.25
0.20
0.15
0.10
0.05
0
25
50
75
100
Degrees of Freedom
method
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db


Mean Bias versus Degrees of Freedom by Model
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db
Mean Bias
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
25
50
75
100
Degrees of Freedom
method
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db